

# BFS Traversal

Step 1: Start at A

## Legend

Complete

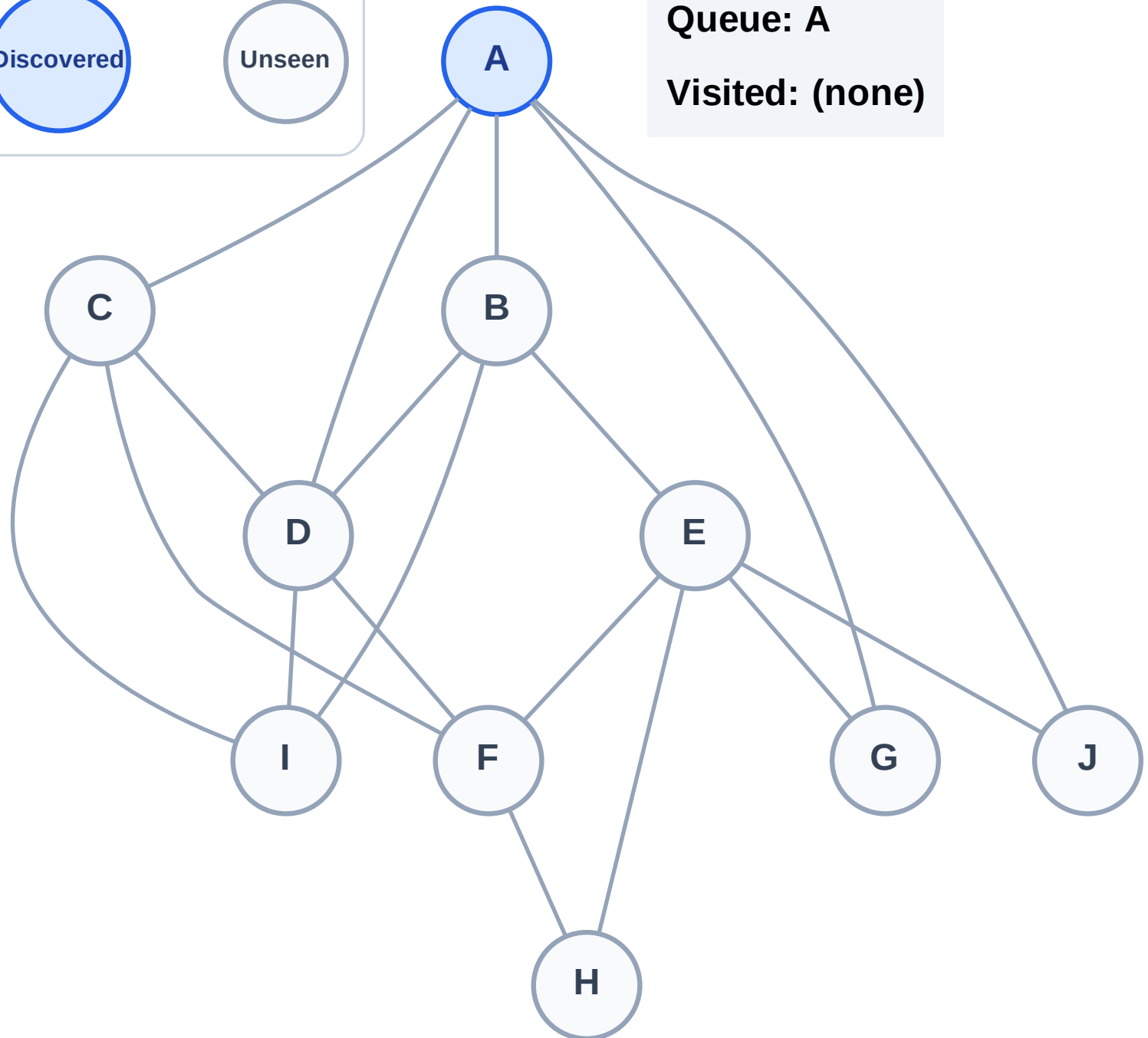
Processing

Discovered

Unseen

Queue: A

Visited: (none)



# BFS Traversal

Step 2: Process node A

## Legend

Complete

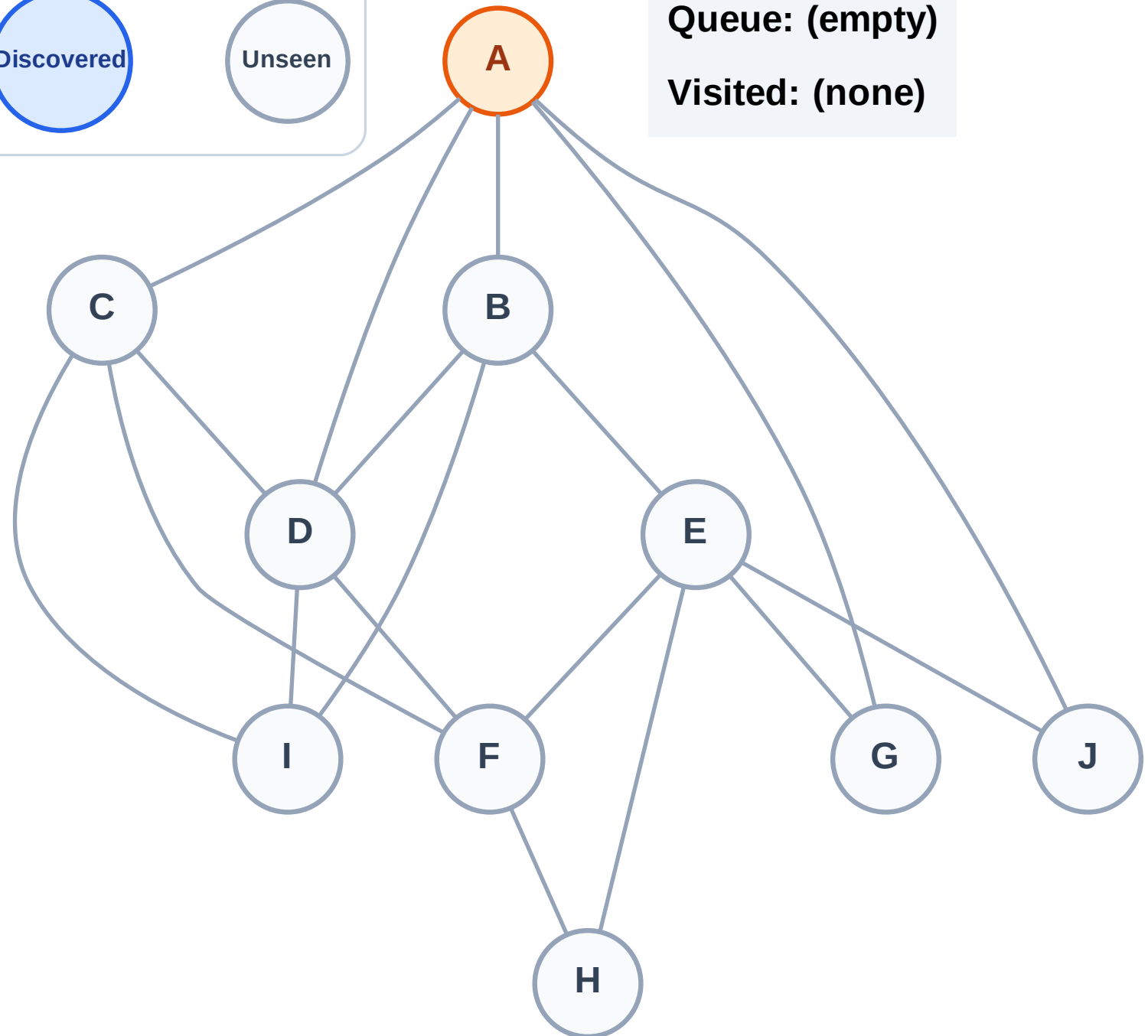
Processing

Discovered

Unseen

Queue: (empty)

Visited: (none)



# BFS Traversal

Step 3: Discover B, C, D, G, J from A

## Legend

Complete

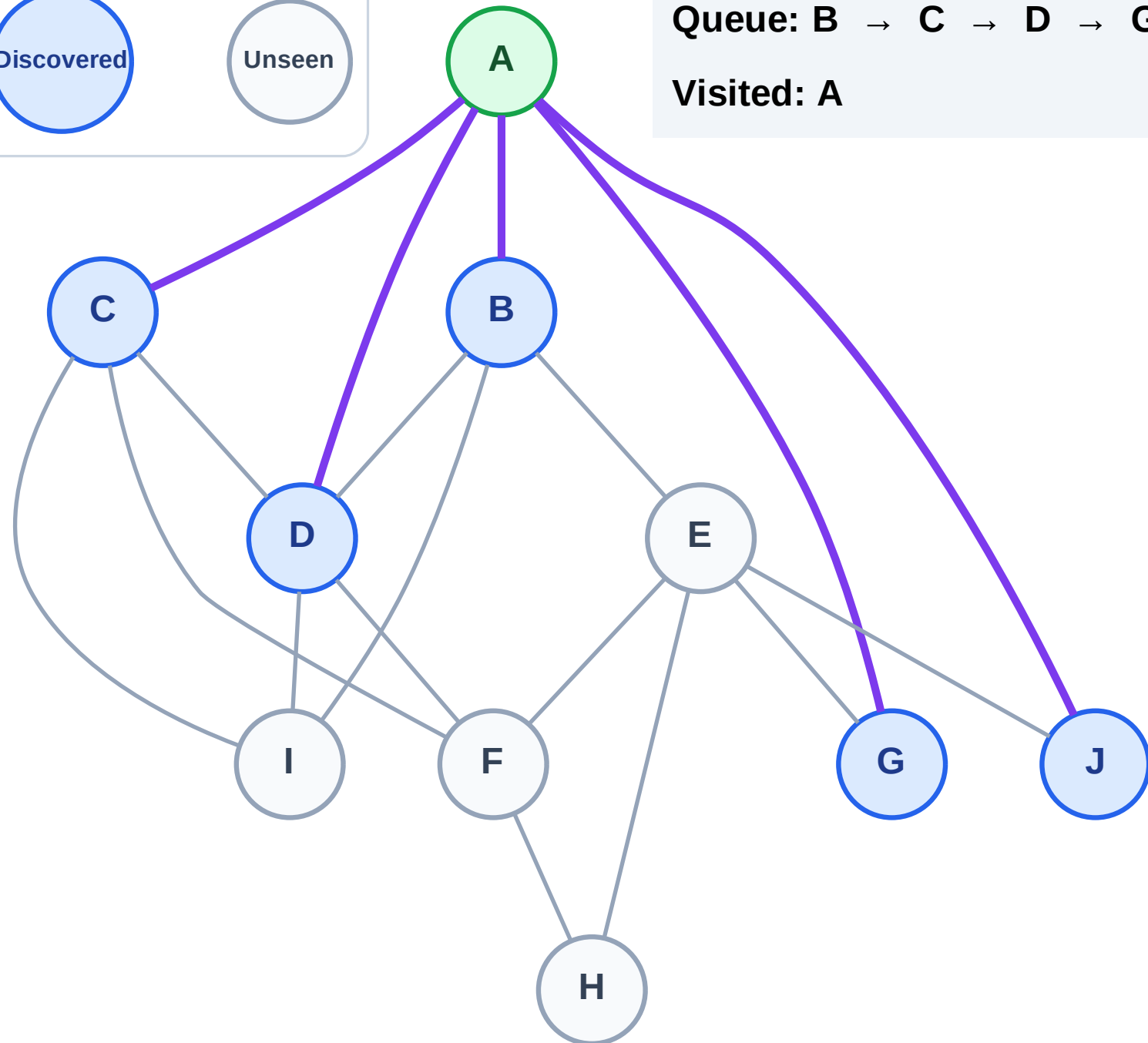
Processing

Discovered

Unseen

Queue: B → C → D → G → J

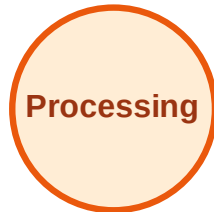
Visited: A



# BFS Traversal

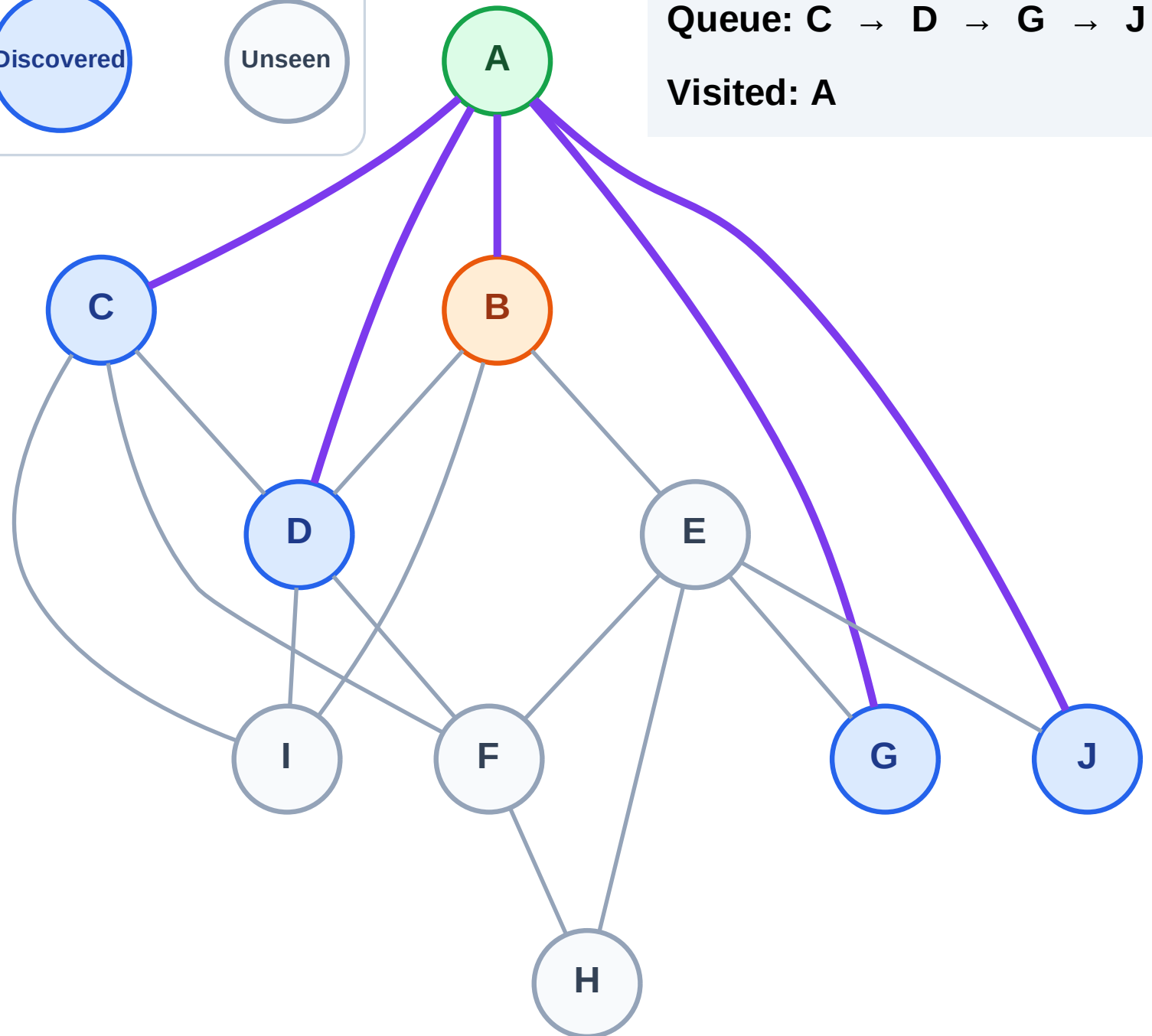
Step 4: Process node B

## Legend



Queue: C → D → G → J

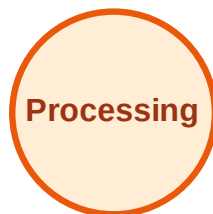
Visited: A



# BFS Traversal

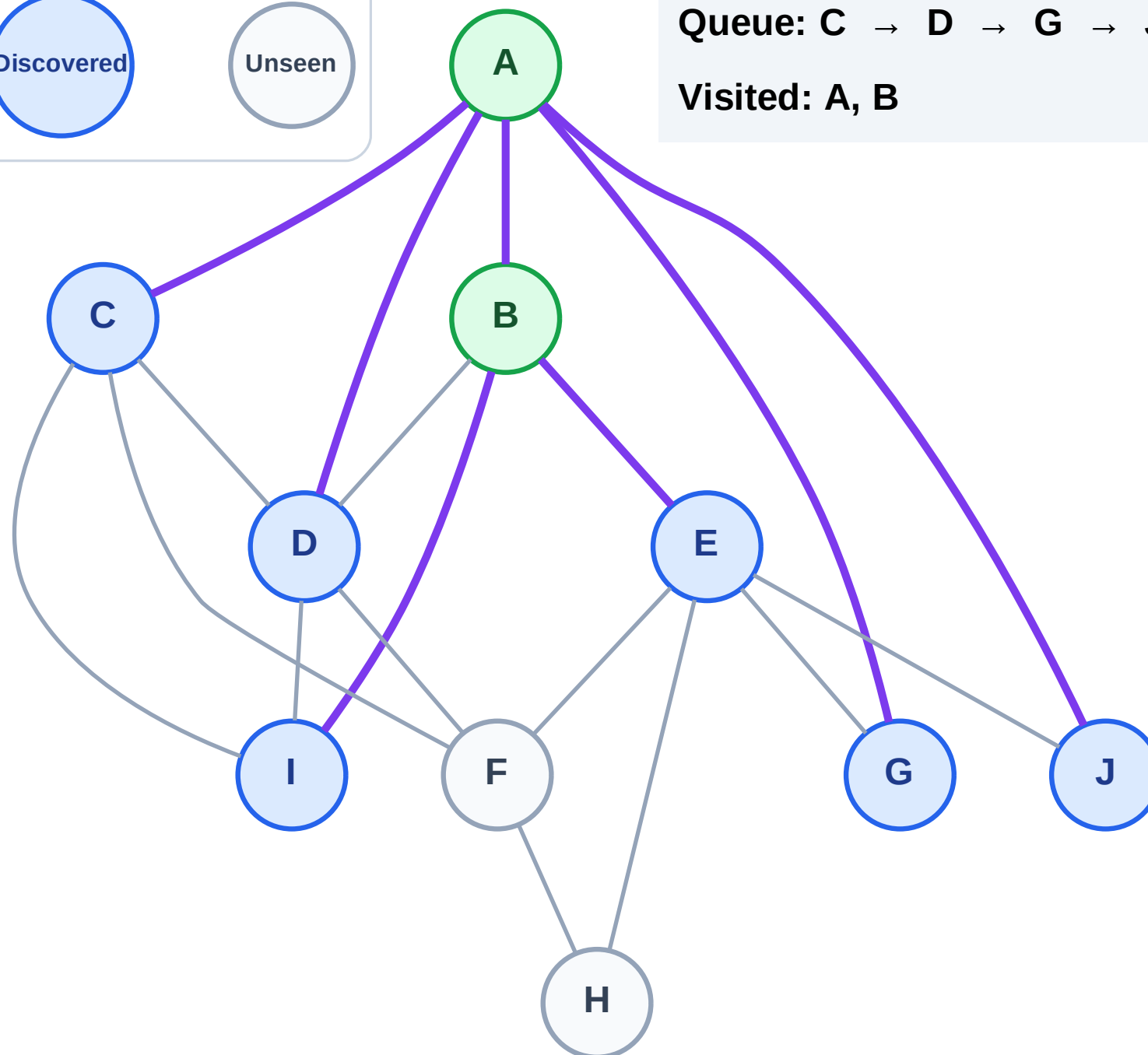
Step 5: Discover E, I from B

## Legend



Queue: C → D → G → J → E → I

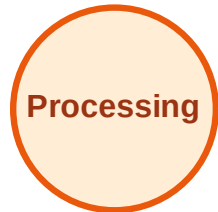
Visited: A, B



# BFS Traversal

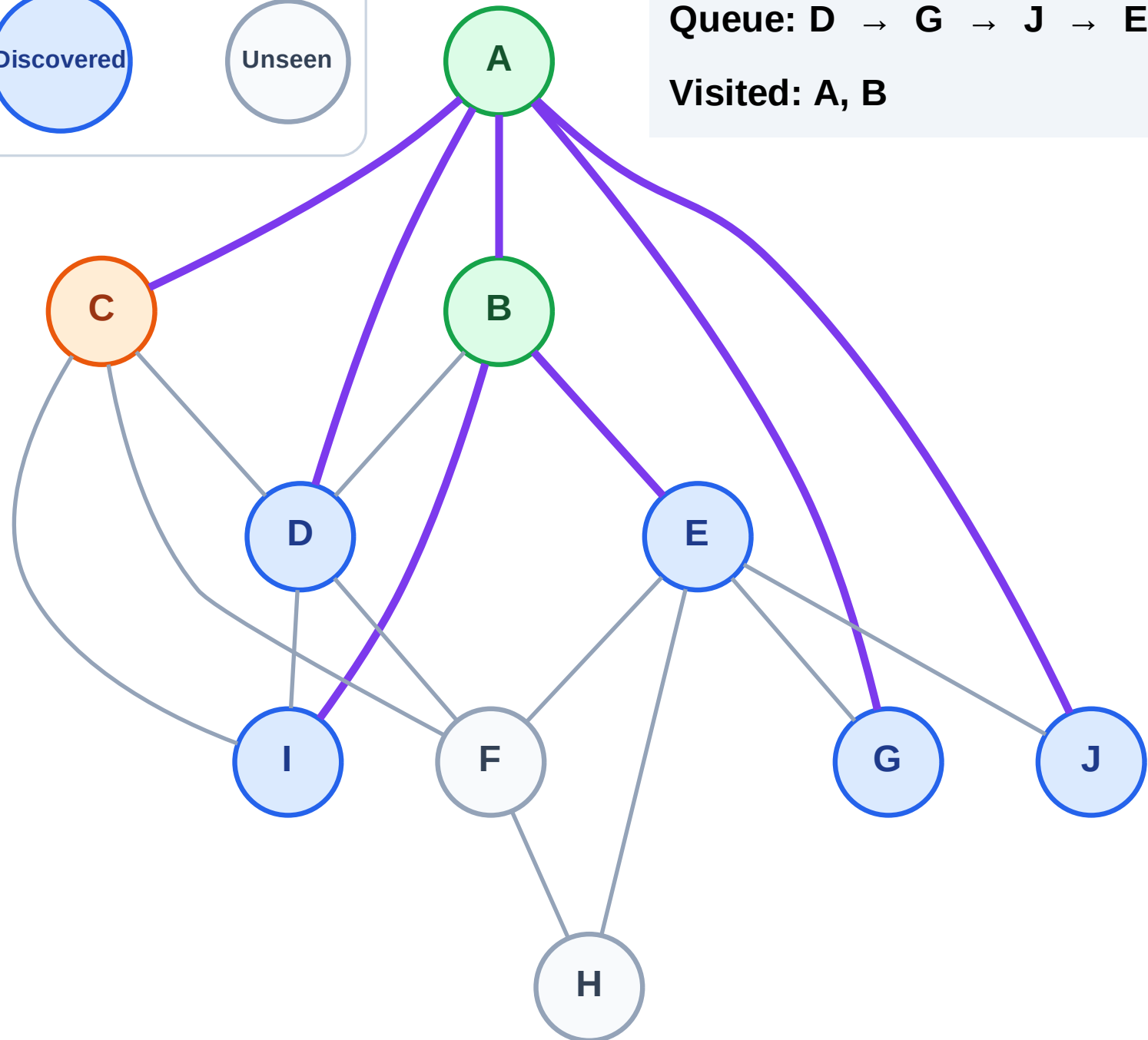
Step 6: Process node C

## Legend



Queue: D → G → J → E → I

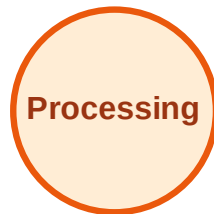
Visited: A, B



# BFS Traversal

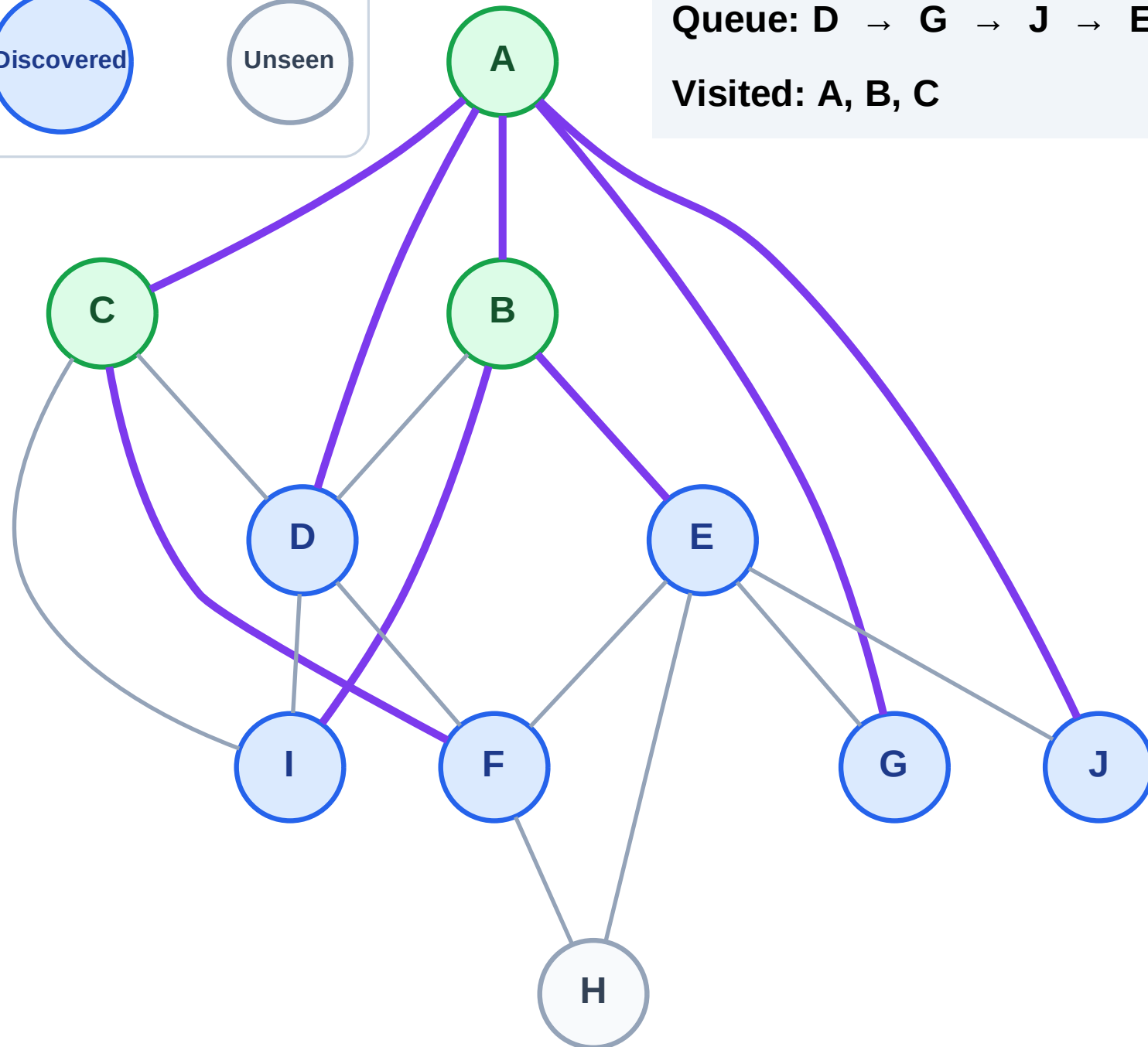
Step 7: Discover F from C

## Legend



Queue: D → G → J → E → I → F

Visited: A, B, C



# BFS Traversal

Step 8: Process node D

## Legend

Complete

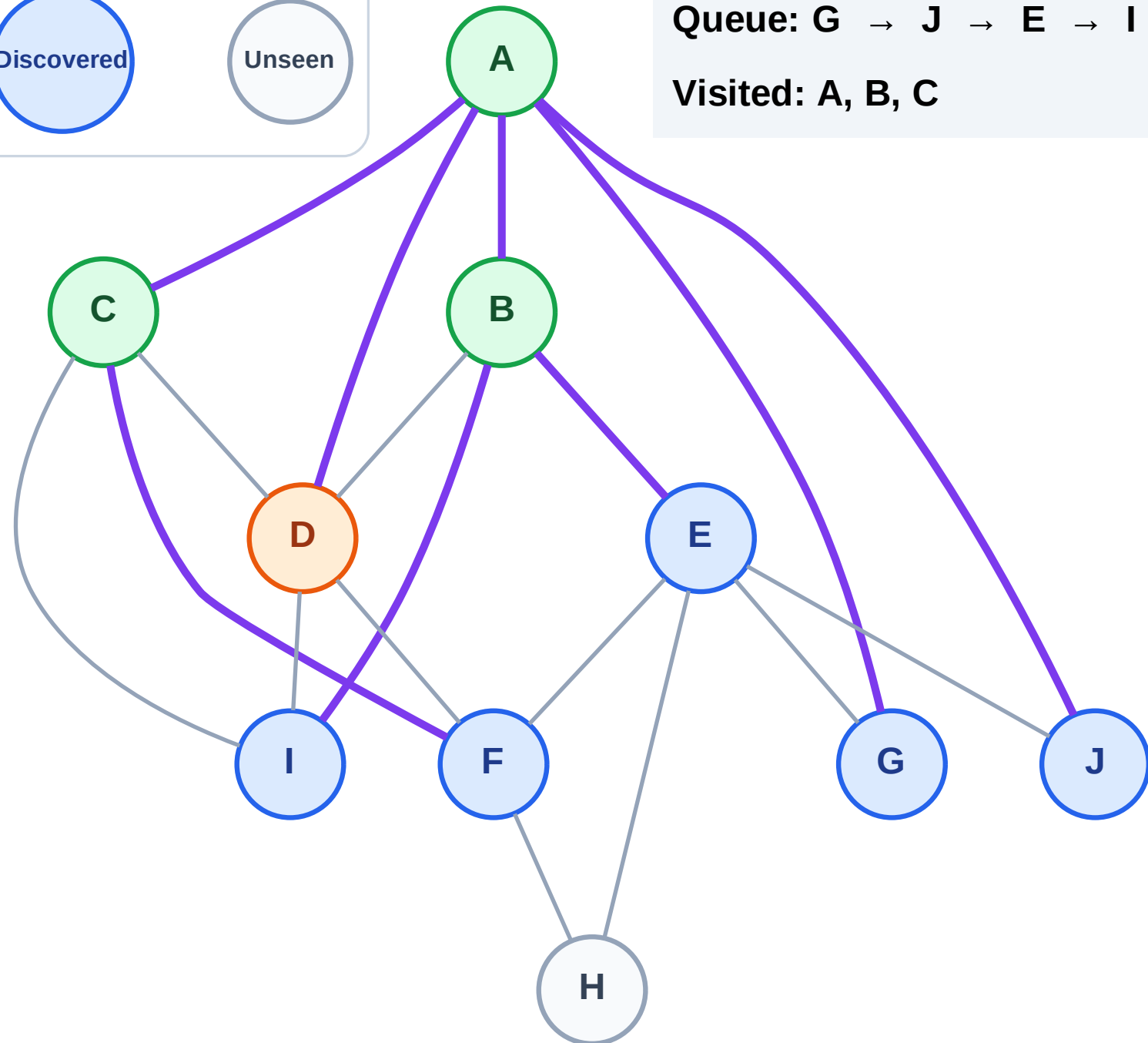
Processing

Discovered

Unseen

Queue: G → J → E → I → F

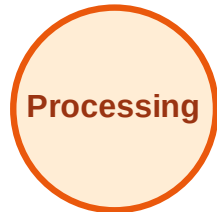
Visited: A, B, C



# BFS Traversal

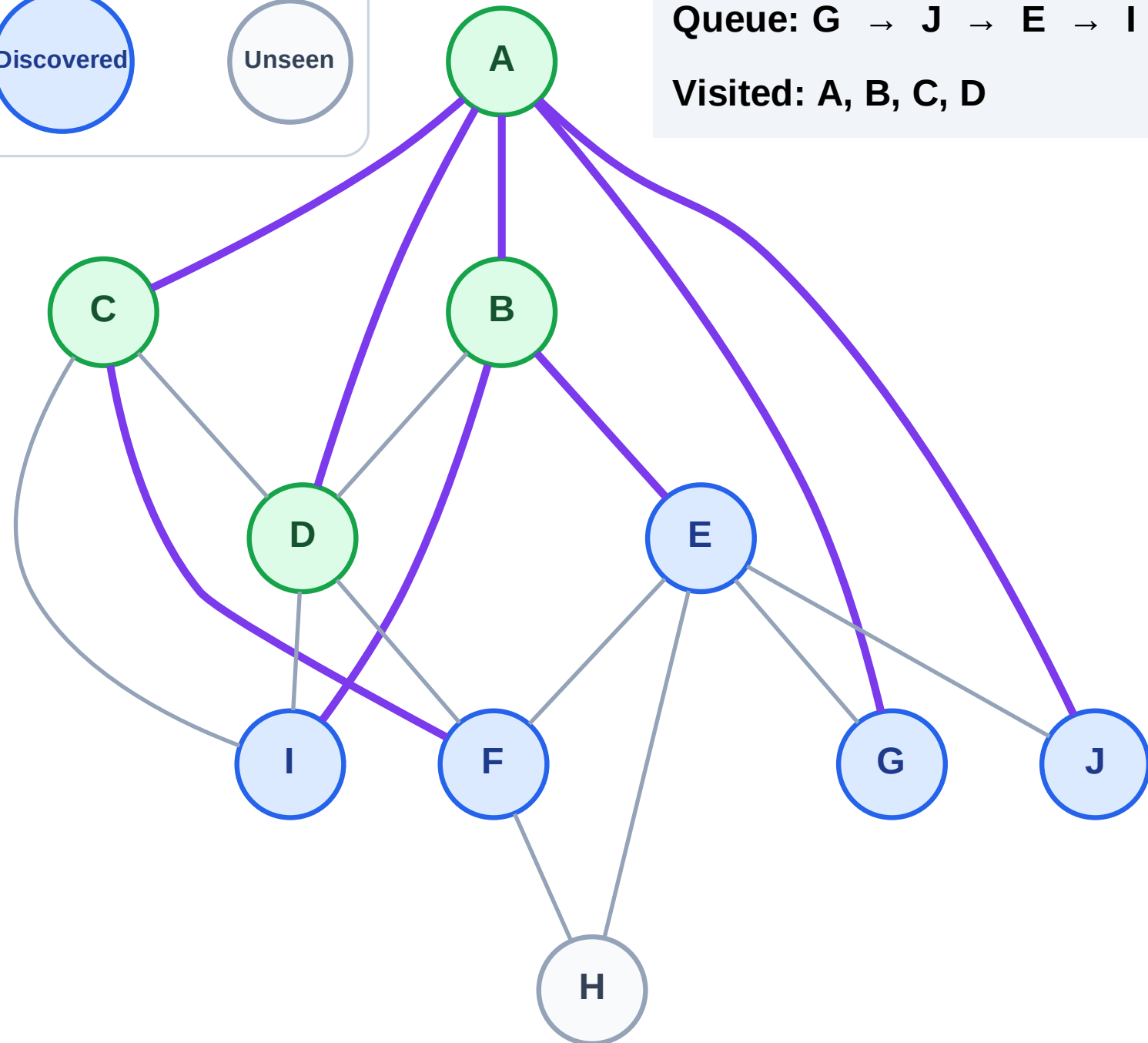
Step 9: No new neighbors from D

## Legend



Queue: G → J → E → I → F

Visited: A, B, C, D



# BFS Traversal

Step 10: Process node G

## Legend

Complete

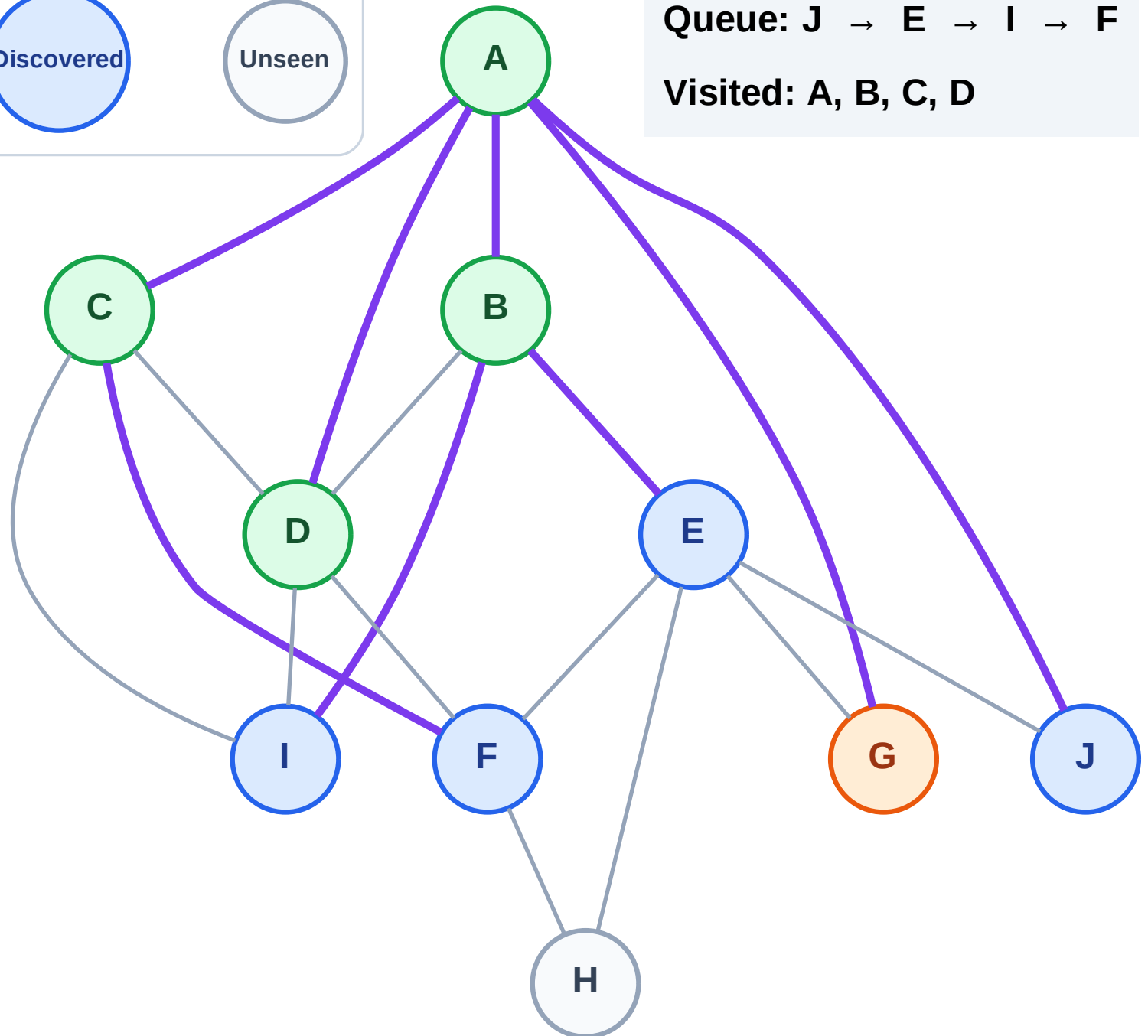
Processing

Discovered

Unseen

Queue: J → E → I → F

Visited: A, B, C, D



# BFS Traversal

Step 11: No new neighbors from G

## Legend

Complete

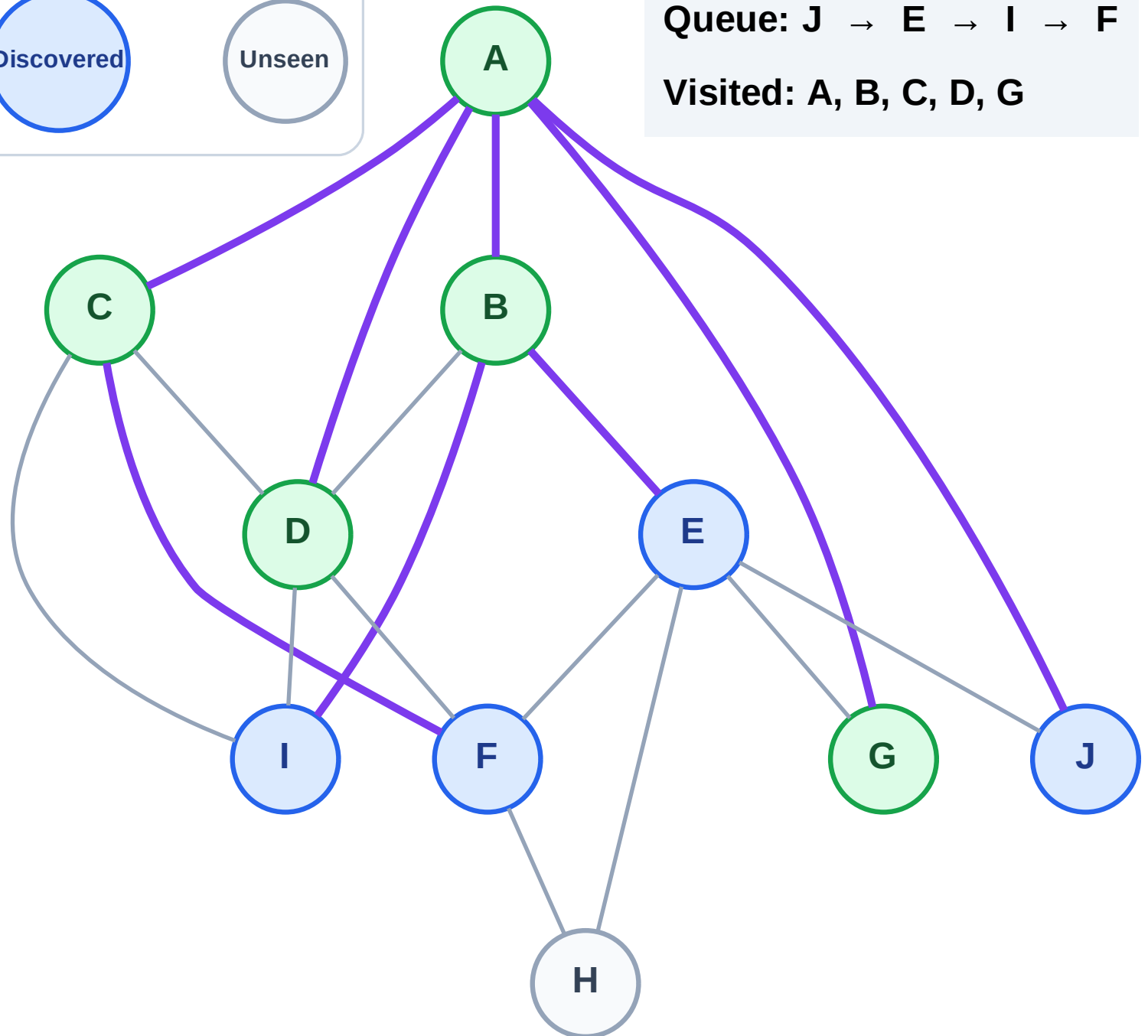
Processing

Discovered

Unseen

Queue: J → E → I → F

Visited: A, B, C, D, G



# BFS Traversal

Step 12: Process node J

## Legend

Complete

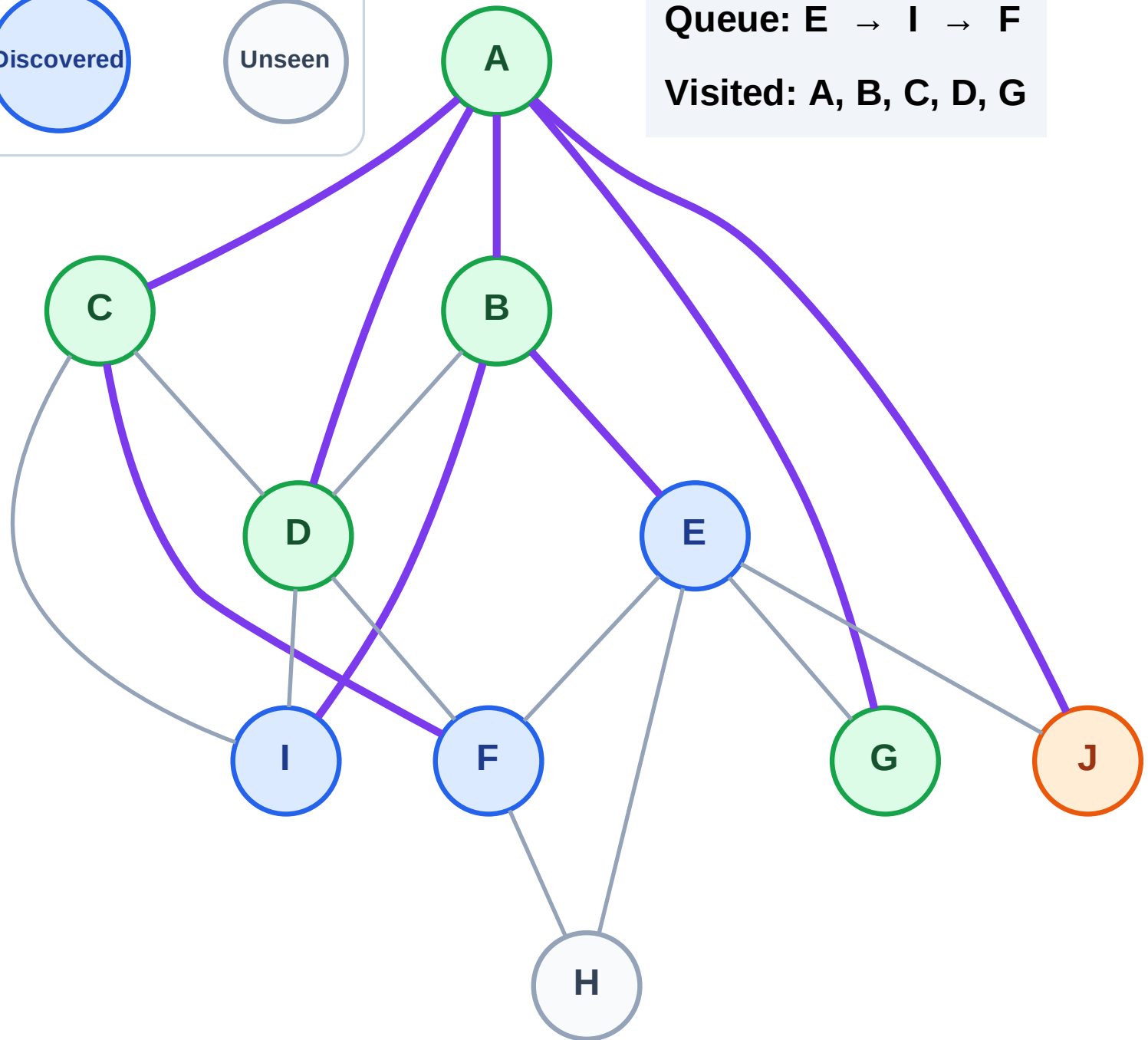
Processing

Discovered

Unseen

Queue: E → I → F

Visited: A, B, C, D, G



# BFS Traversal

Step 13: No new neighbors from J

## Legend

Complete

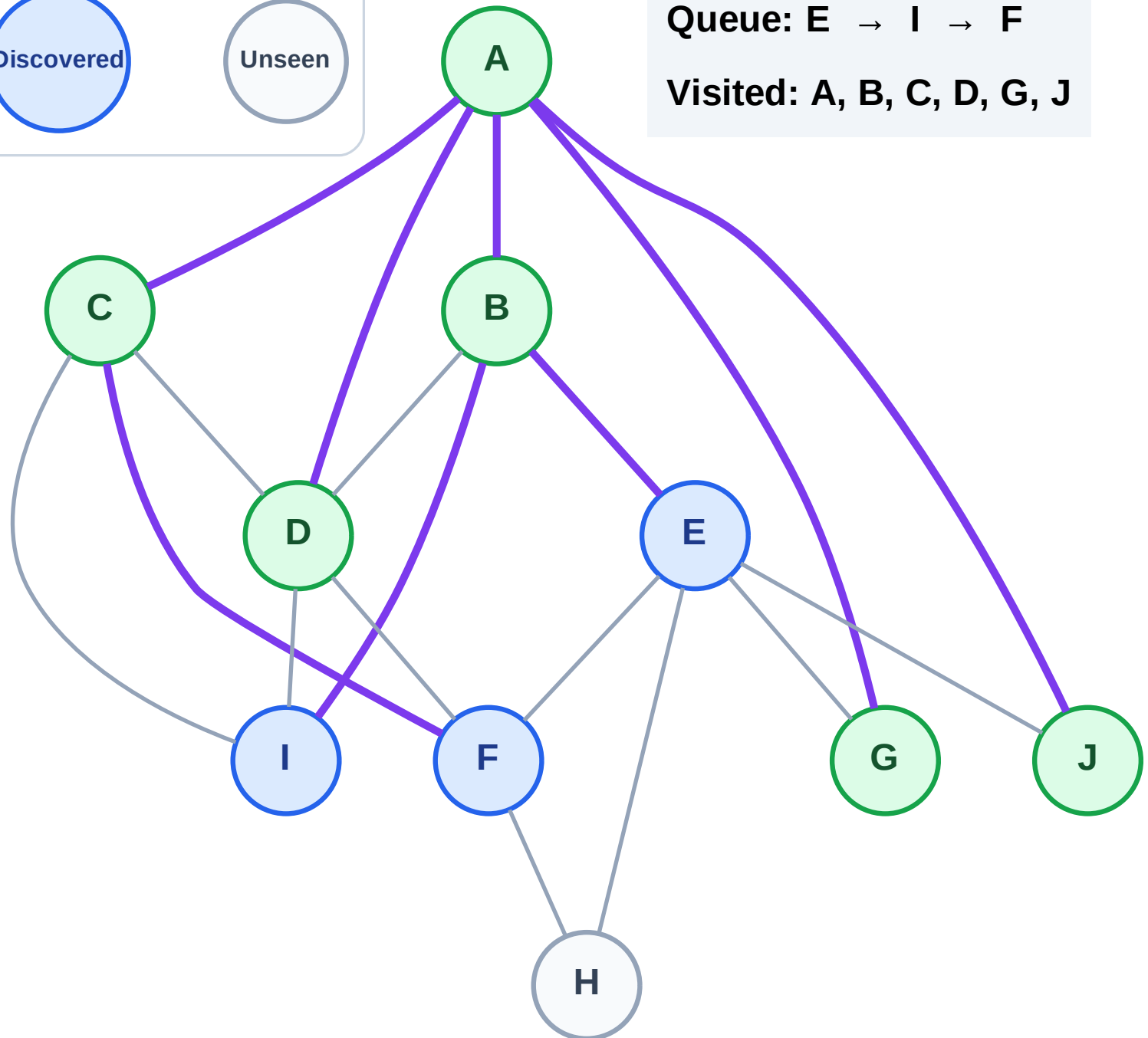
Processing

Discovered

Unseen

Queue: E → I → F

Visited: A, B, C, D, G, J



# BFS Traversal

Step 14: Process node E

## Legend

Complete

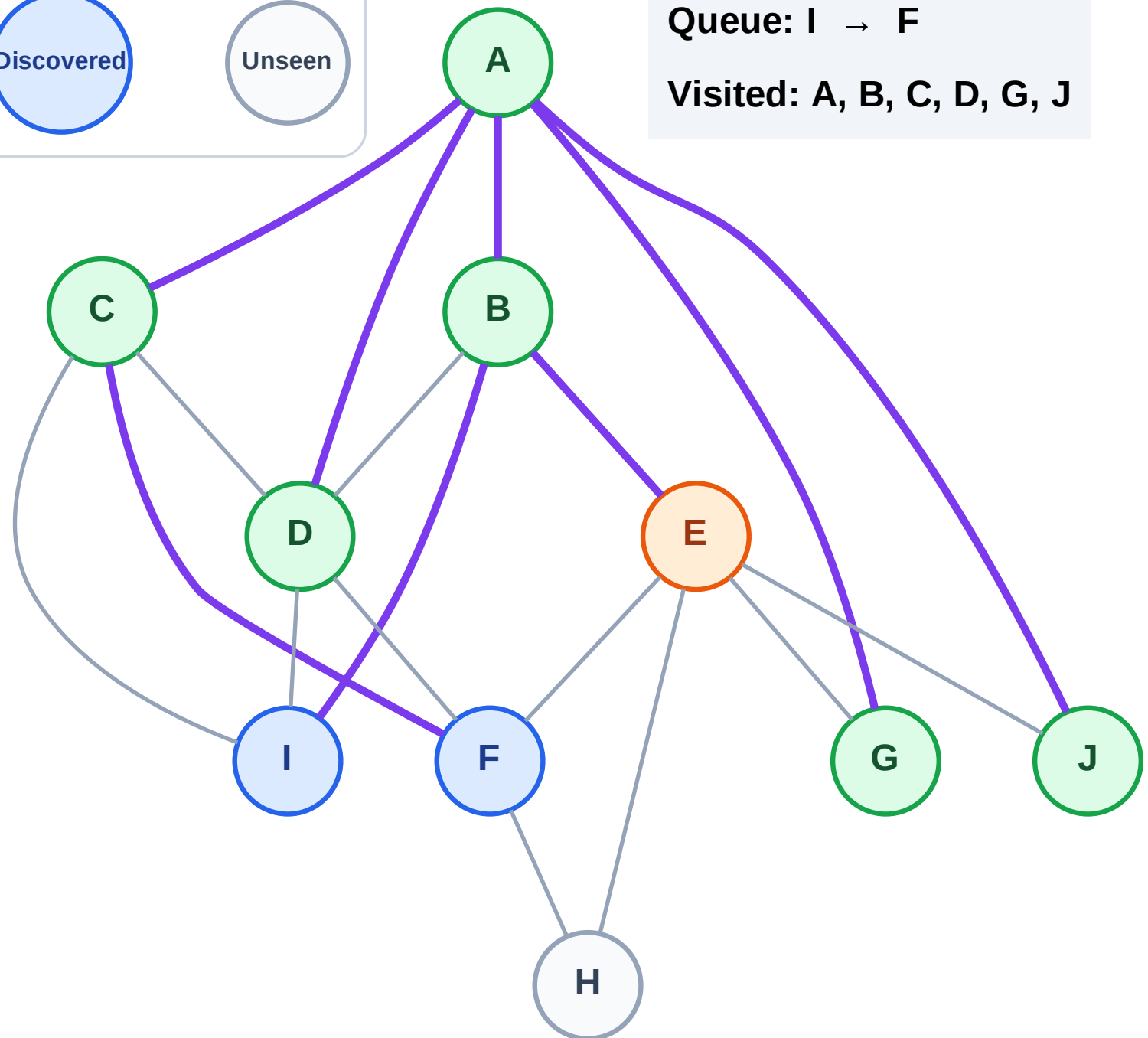
Processing

Discovered

Unseen

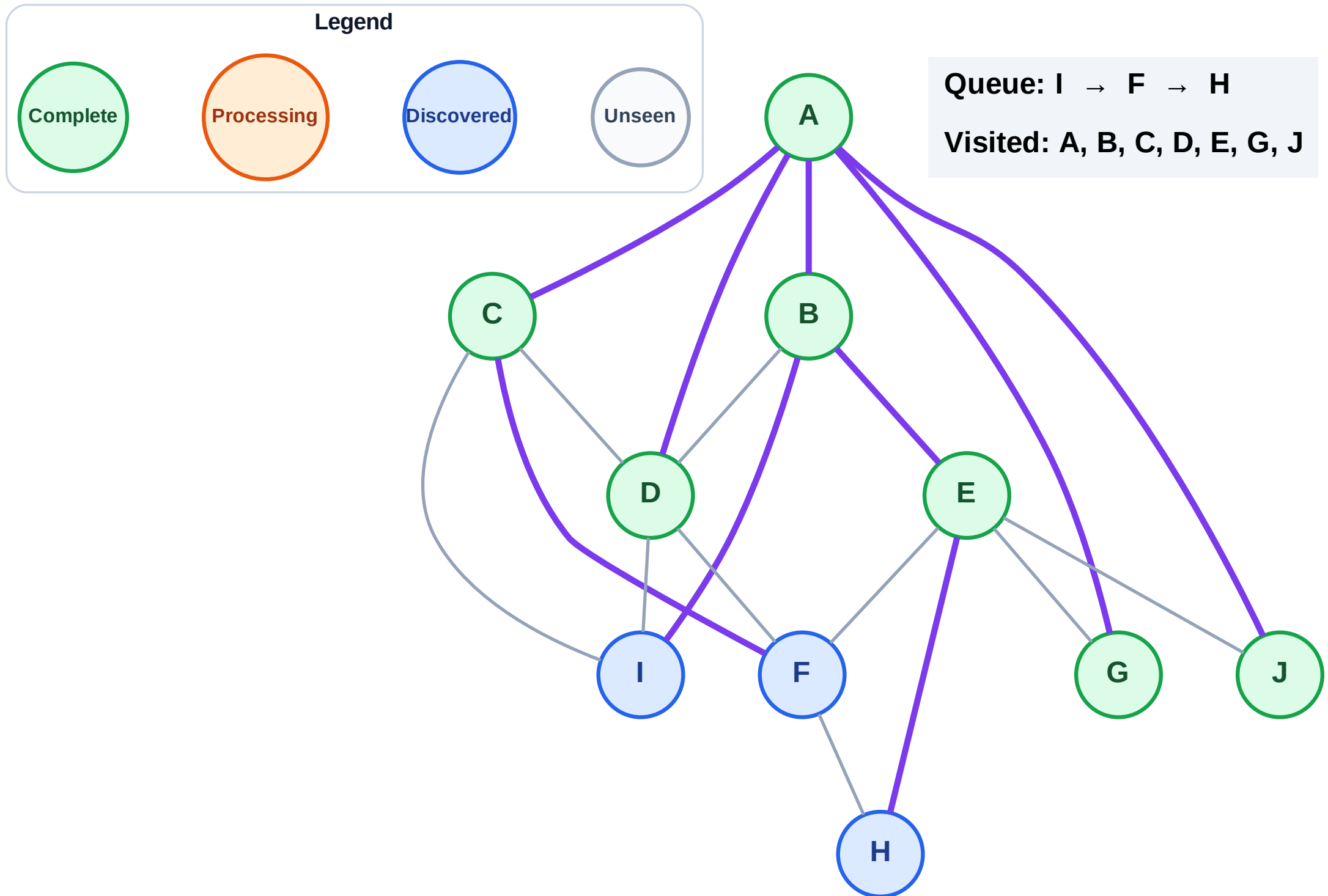
Queue: I → F

Visited: A, B, C, D, G, J



## Step 15: Discover H from E

## Step 15: Discover H from E



# BFS Traversal

Step 16: Process node I

## Legend

Complete

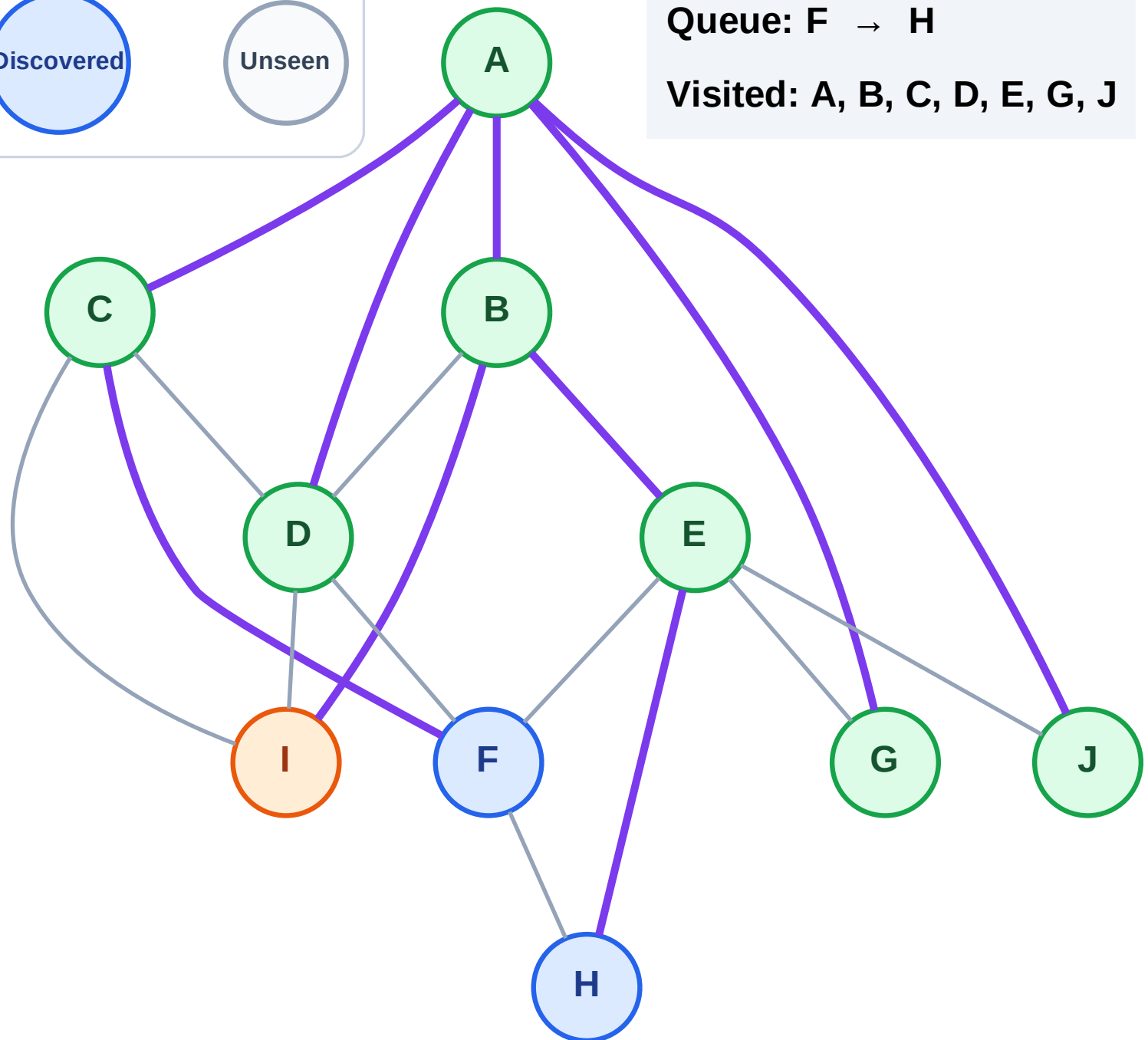
Processing

Discovered

Unseen

Queue: F → H

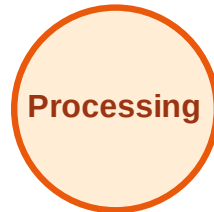
Visited: A, B, C, D, E, G, J



# BFS Traversal

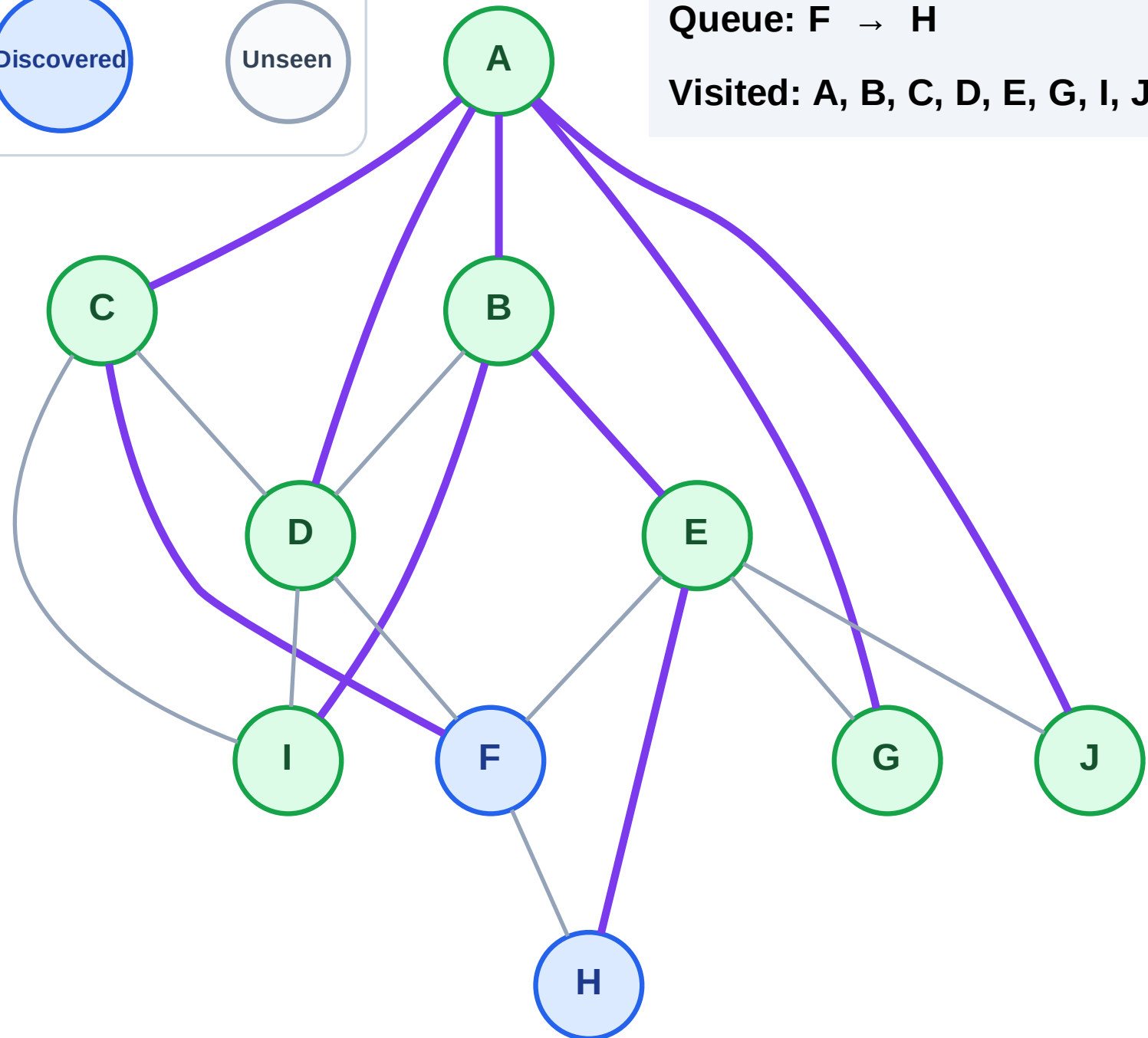
Step 17: No new neighbors from I

## Legend



Queue: F → H

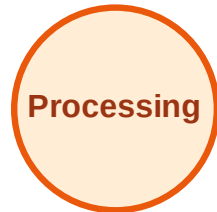
Visited: A, B, C, D, E, G, I, J



# BFS Traversal

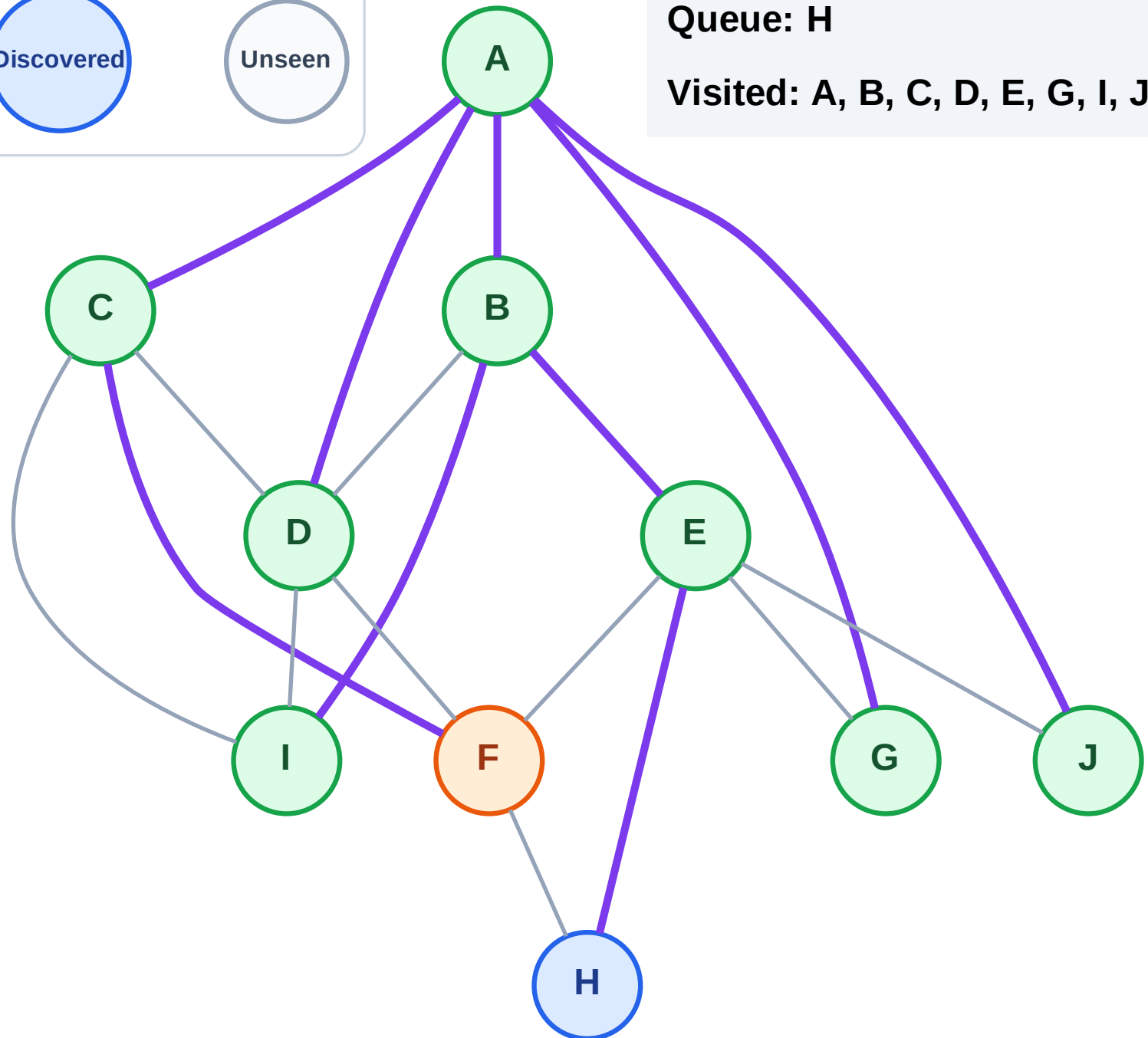
Step 18: Process node F

## Legend



Queue: H

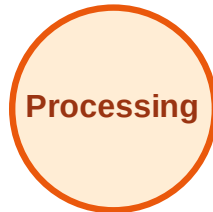
Visited: A, B, C, D, E, G, I, J



# BFS Traversal

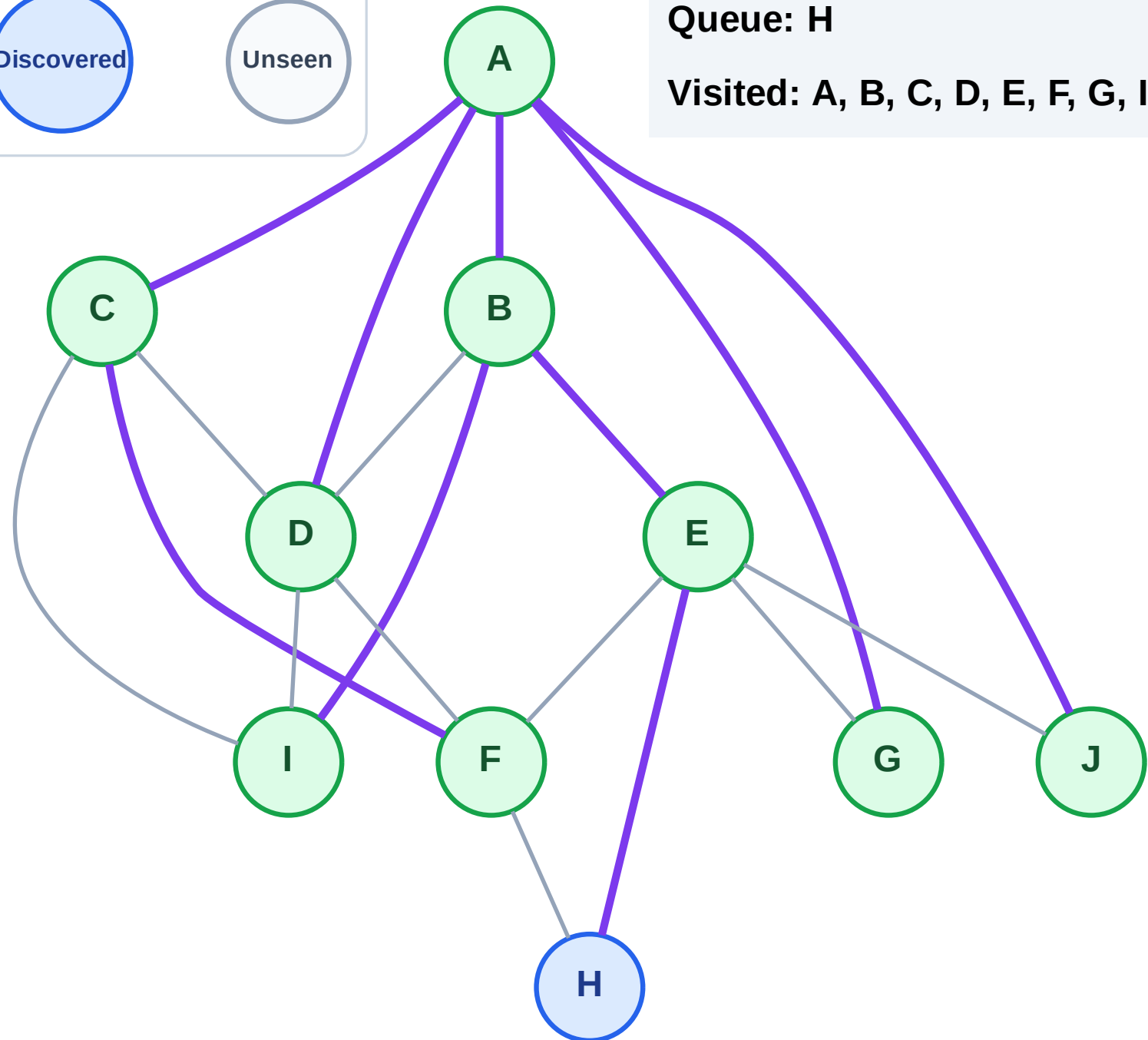
Step 19: No new neighbors from F

## Legend



Queue: H

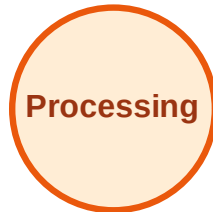
Visited: A, B, C, D, E, F, G, I, J



# BFS Traversal

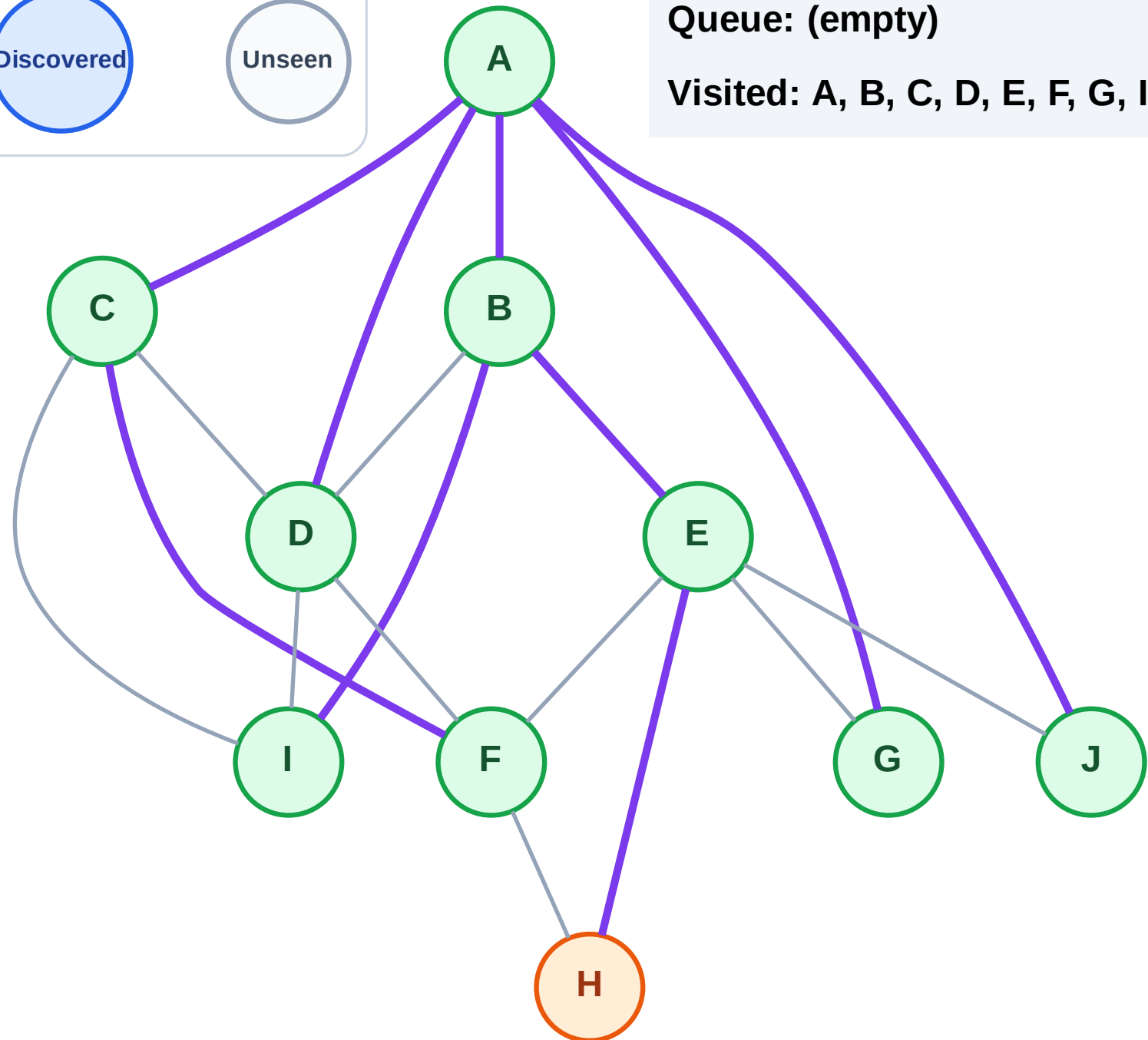
Step 20: Process node H

## Legend



Queue: (empty)

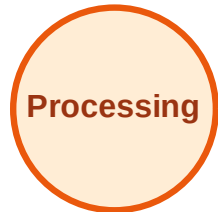
Visited: A, B, C, D, E, F, G, I, J



# BFS Traversal

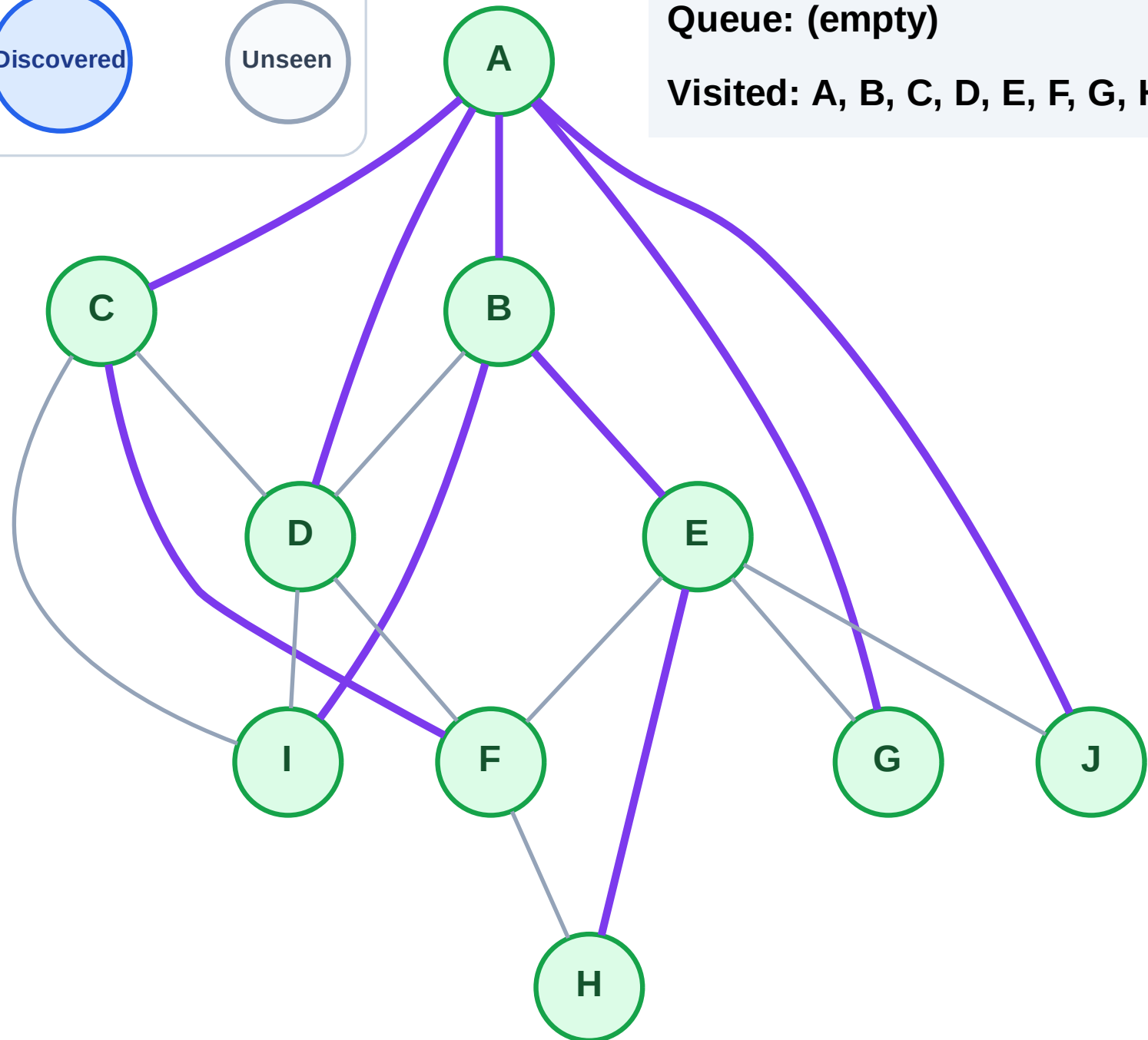
Step 21: No new neighbors from H

## Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

Step 1: Start at A

## Legend

Complete

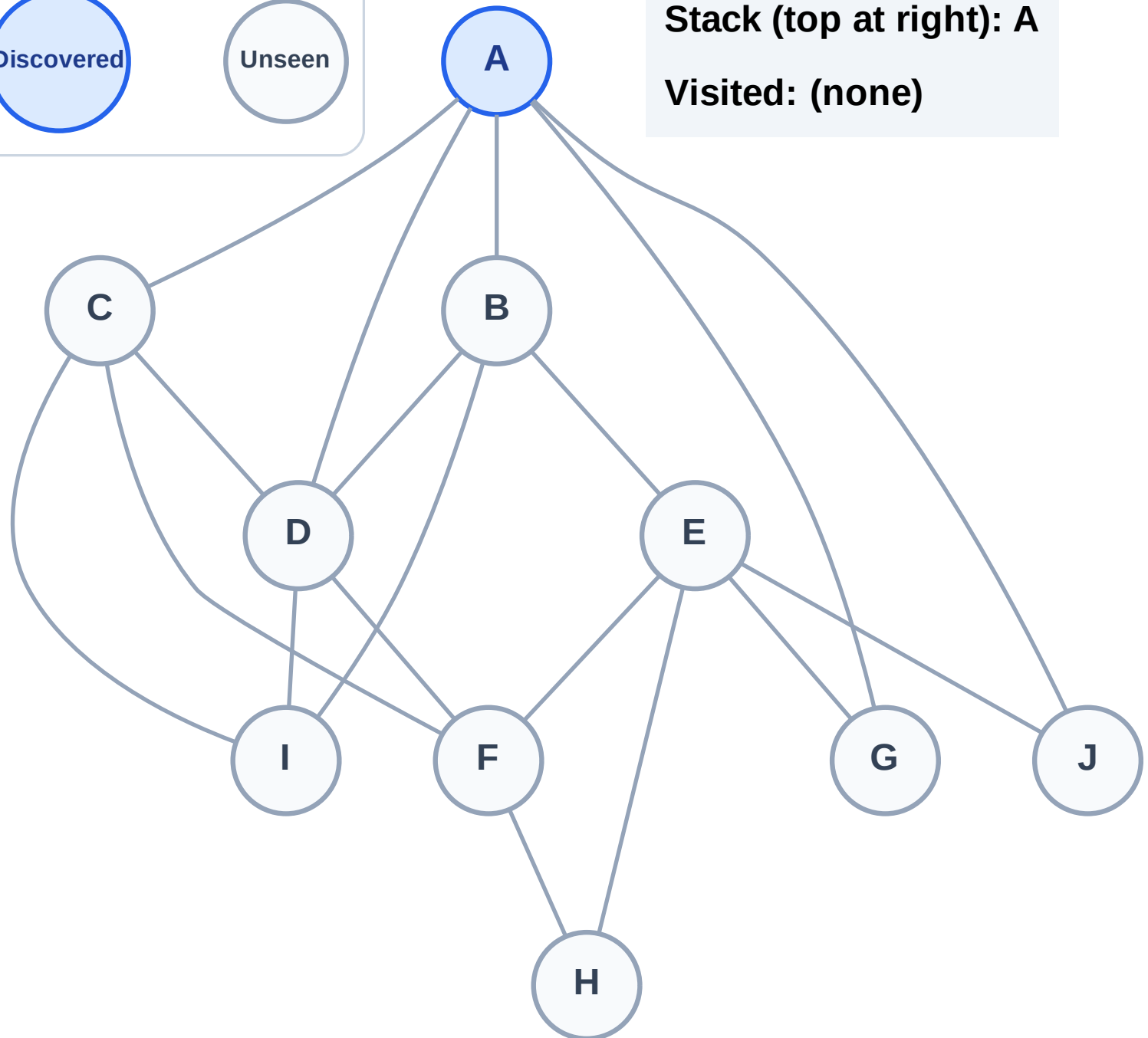
Processing

Discovered

Unseen

Stack (top at right): A

Visited: (none)



# DFS Traversal

Step 2: Process node A

## Legend

Complete

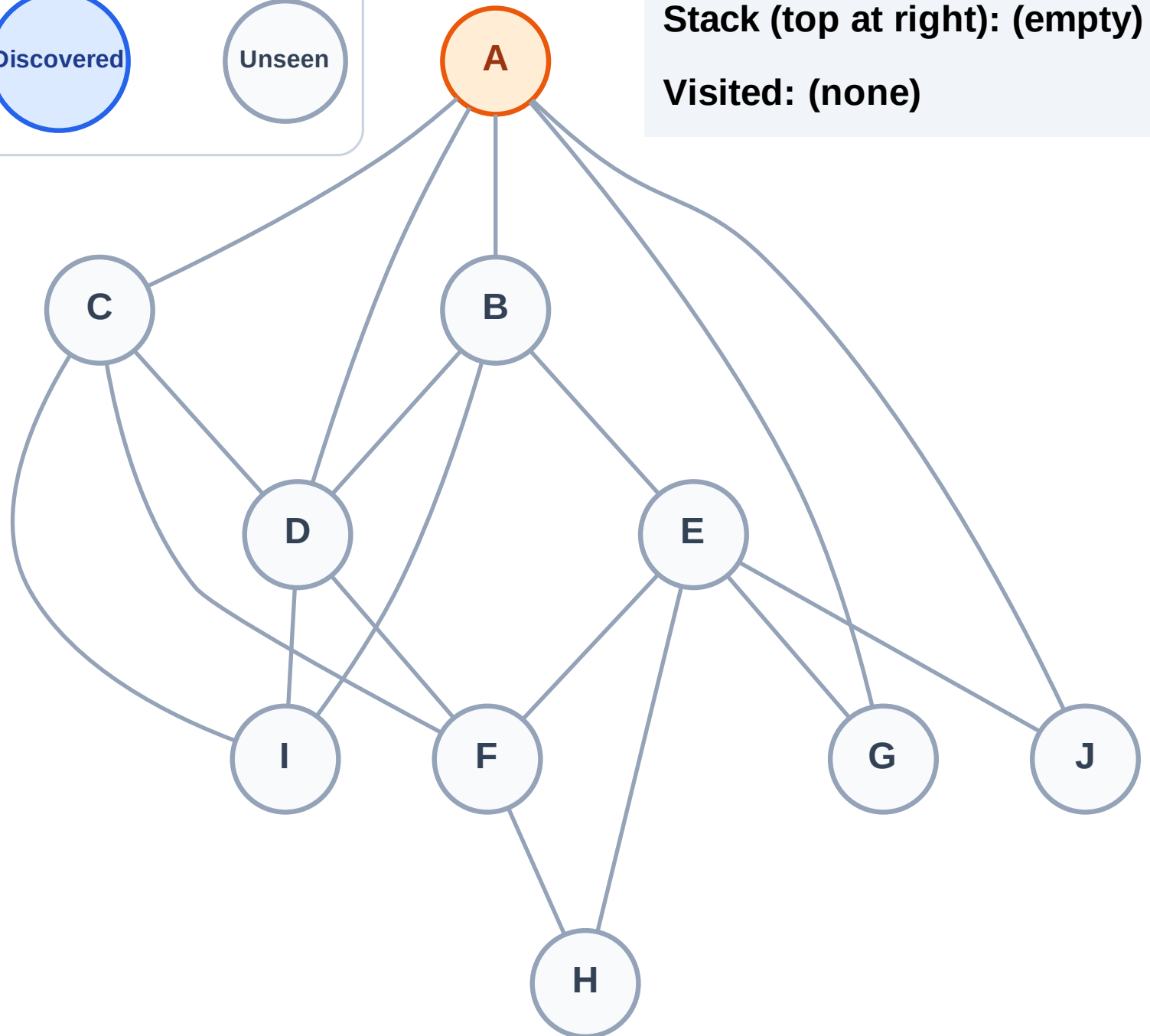
Processing

Discovered

Unseen

Stack (top at right): (empty)

Visited: (none)



# DFS Traversal

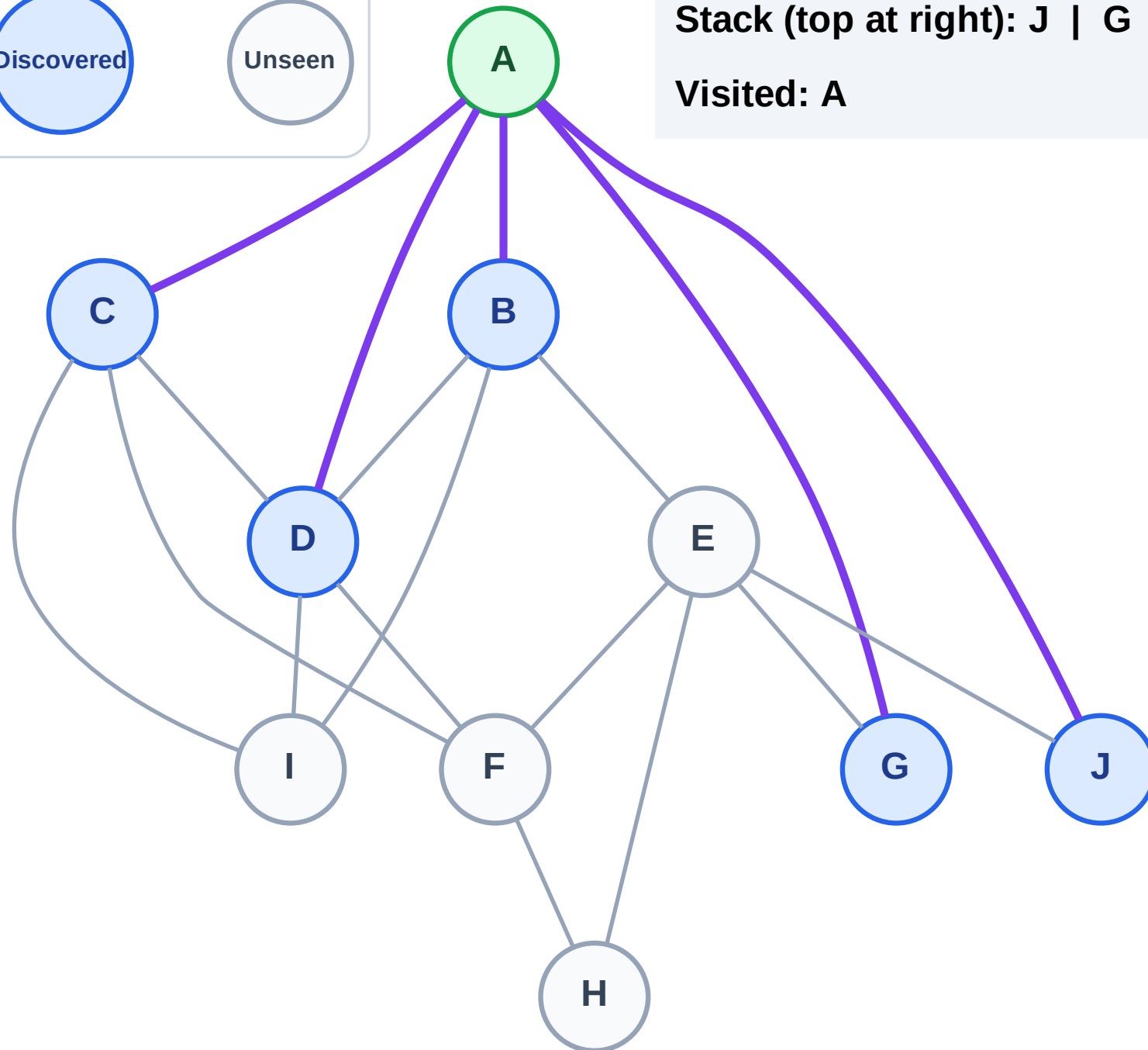
Step 3: Discover J, G, D, C, B from A

## Legend



Stack (top at right): J | G | D | C | B

Visited: A



# DFS Traversal

Step 4: Process node B

Legend

Complete

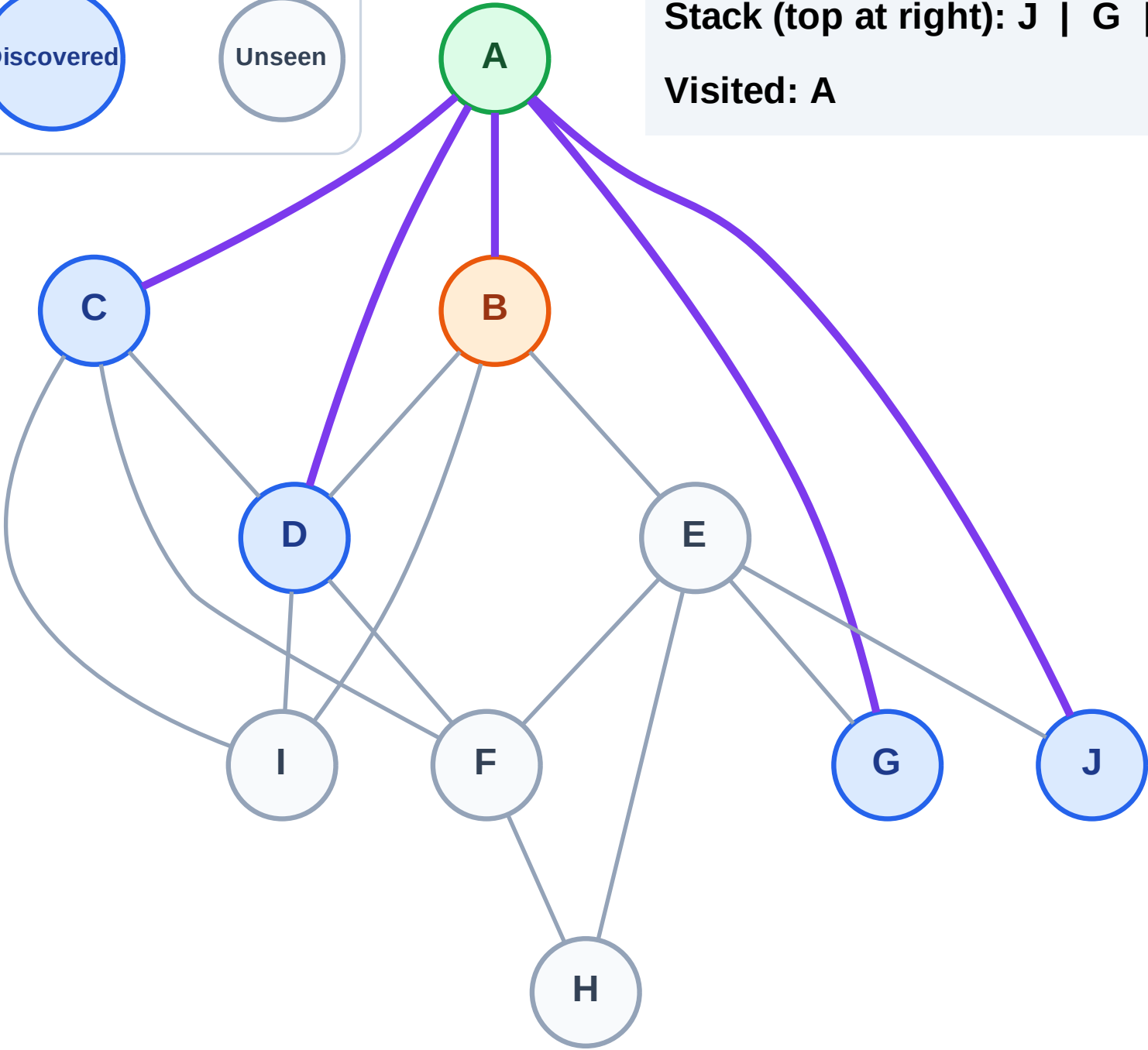
Processing

Discovered

Unseen

Stack (top at right): J | G | D | C

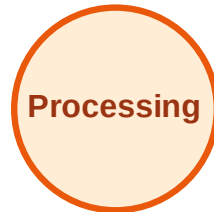
Visited: A



# DFS Traversal

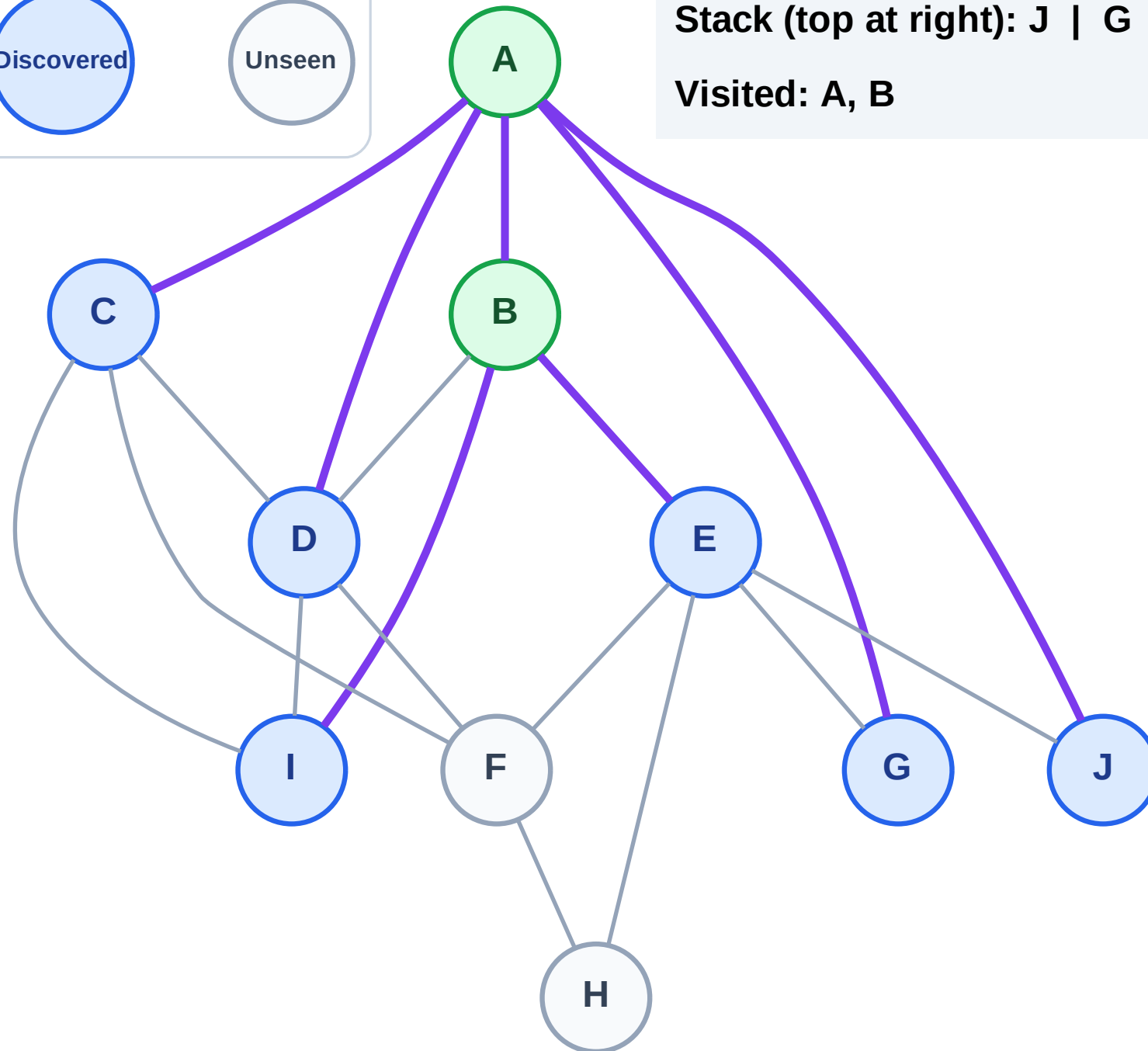
Step 5: Discover I, E from B

## Legend



Stack (top at right): J | G | D | C | I | E

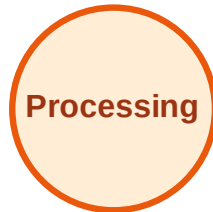
Visited: A, B



# DFS Traversal

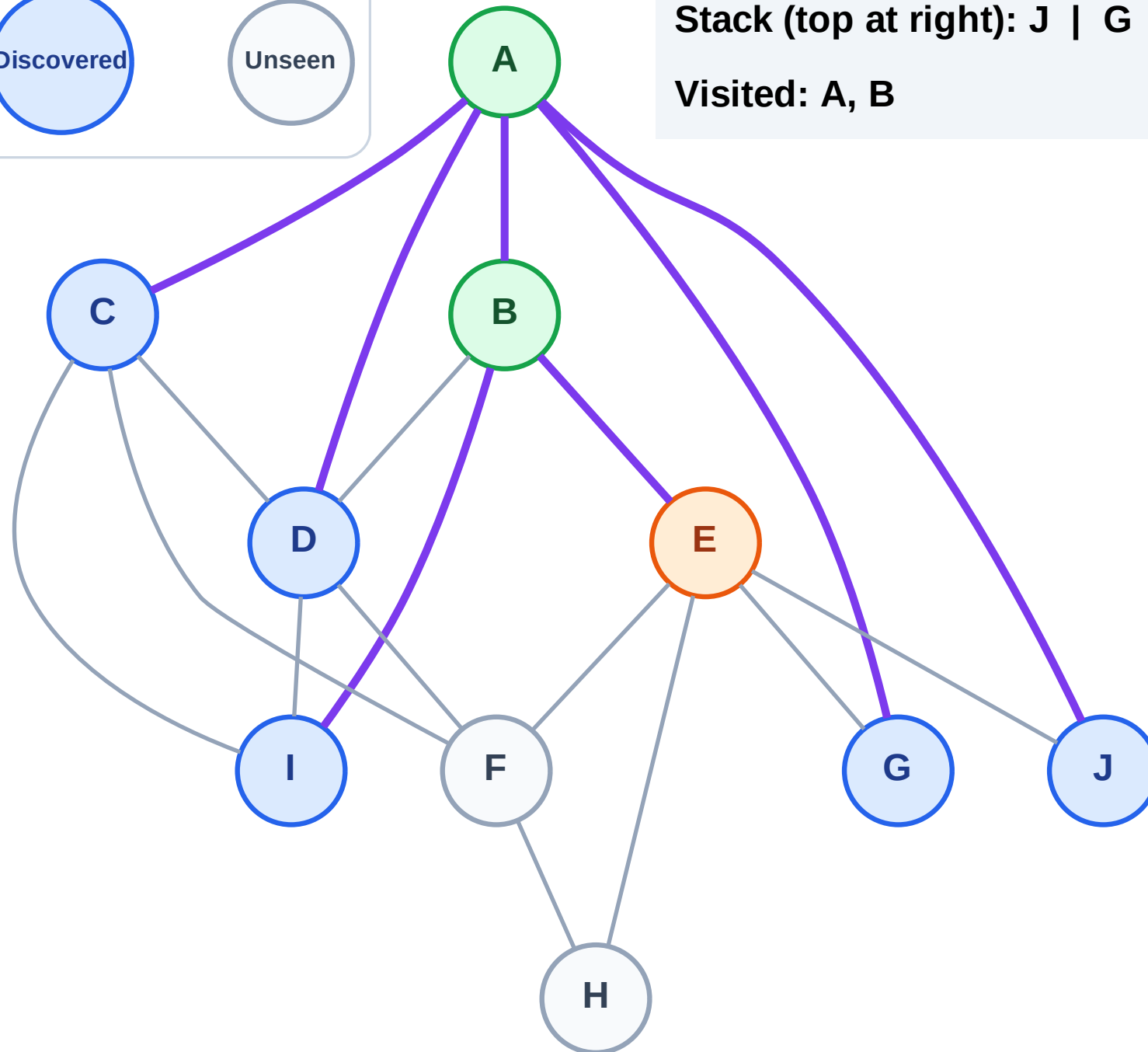
Step 6: Process node E

## Legend



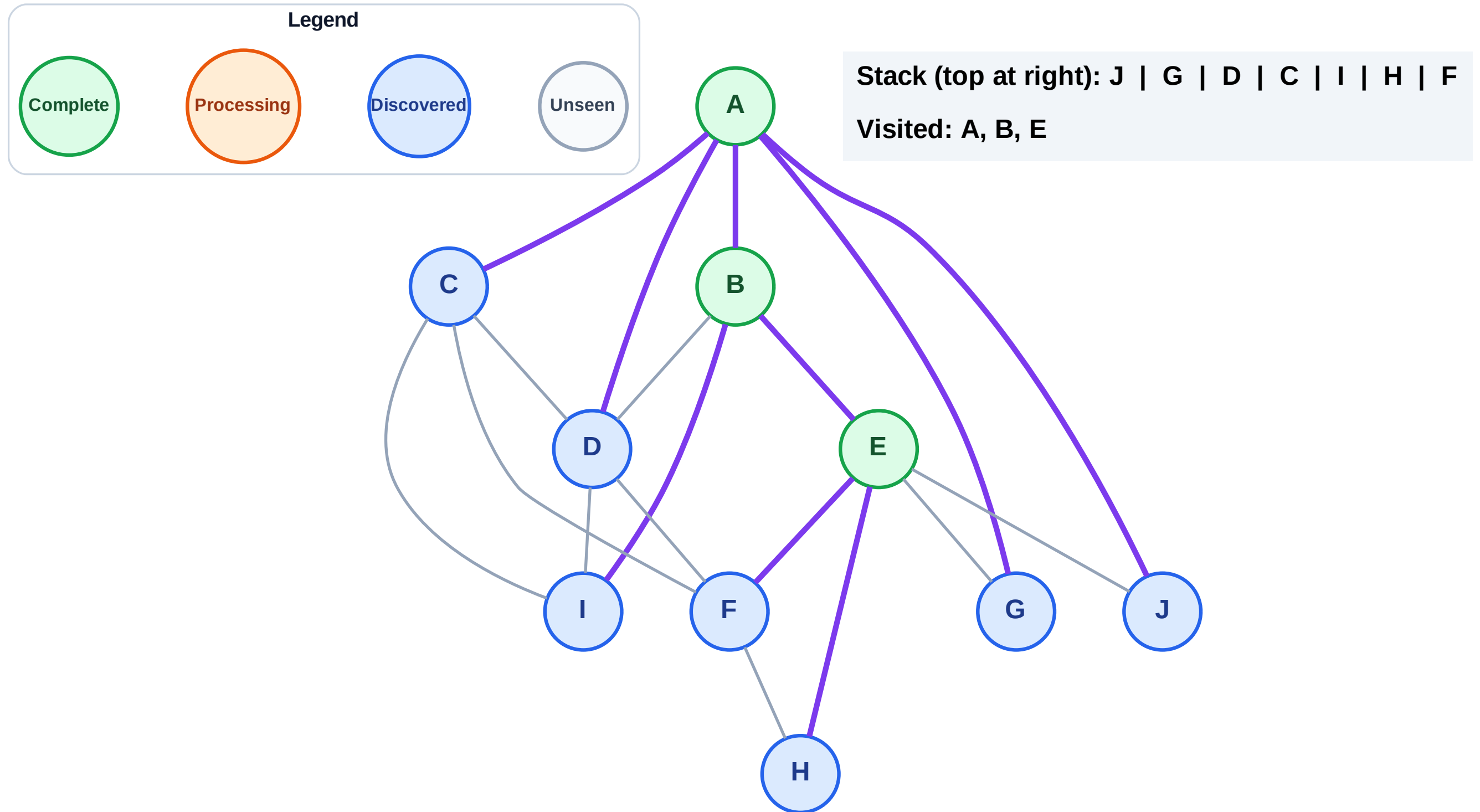
Stack (top at right): J | G | D | C | I

Visited: A, B



### Step 7: Discover H, F from E

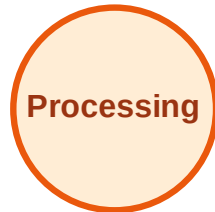
### Step 7: Discover H, F from E



# DFS Traversal

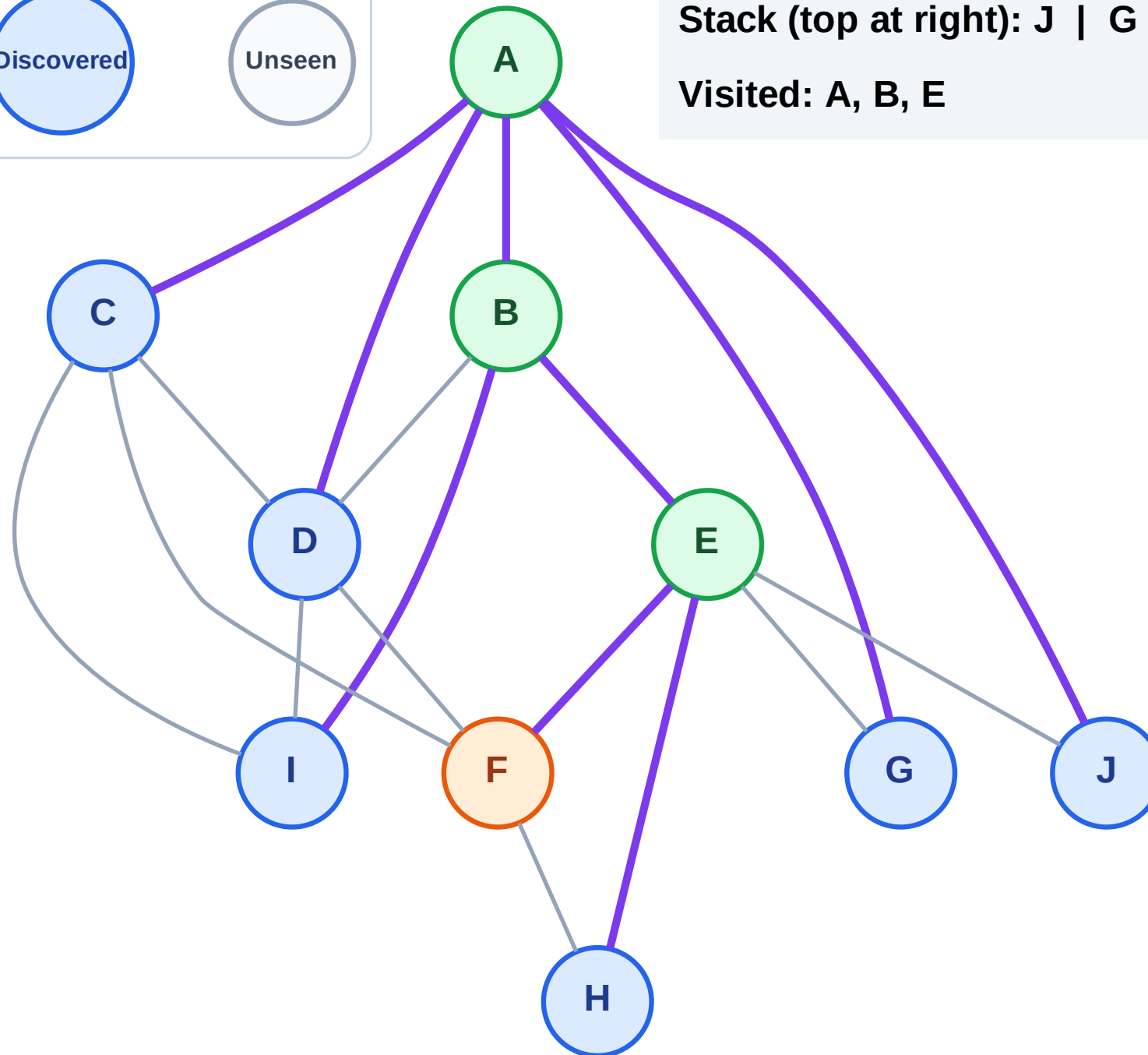
Step 8: Process node F

## Legend



Stack (top at right): J | G | D | C | I | H

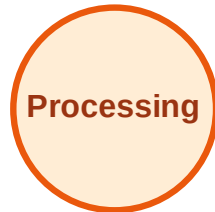
Visited: A, B, E



# DFS Traversal

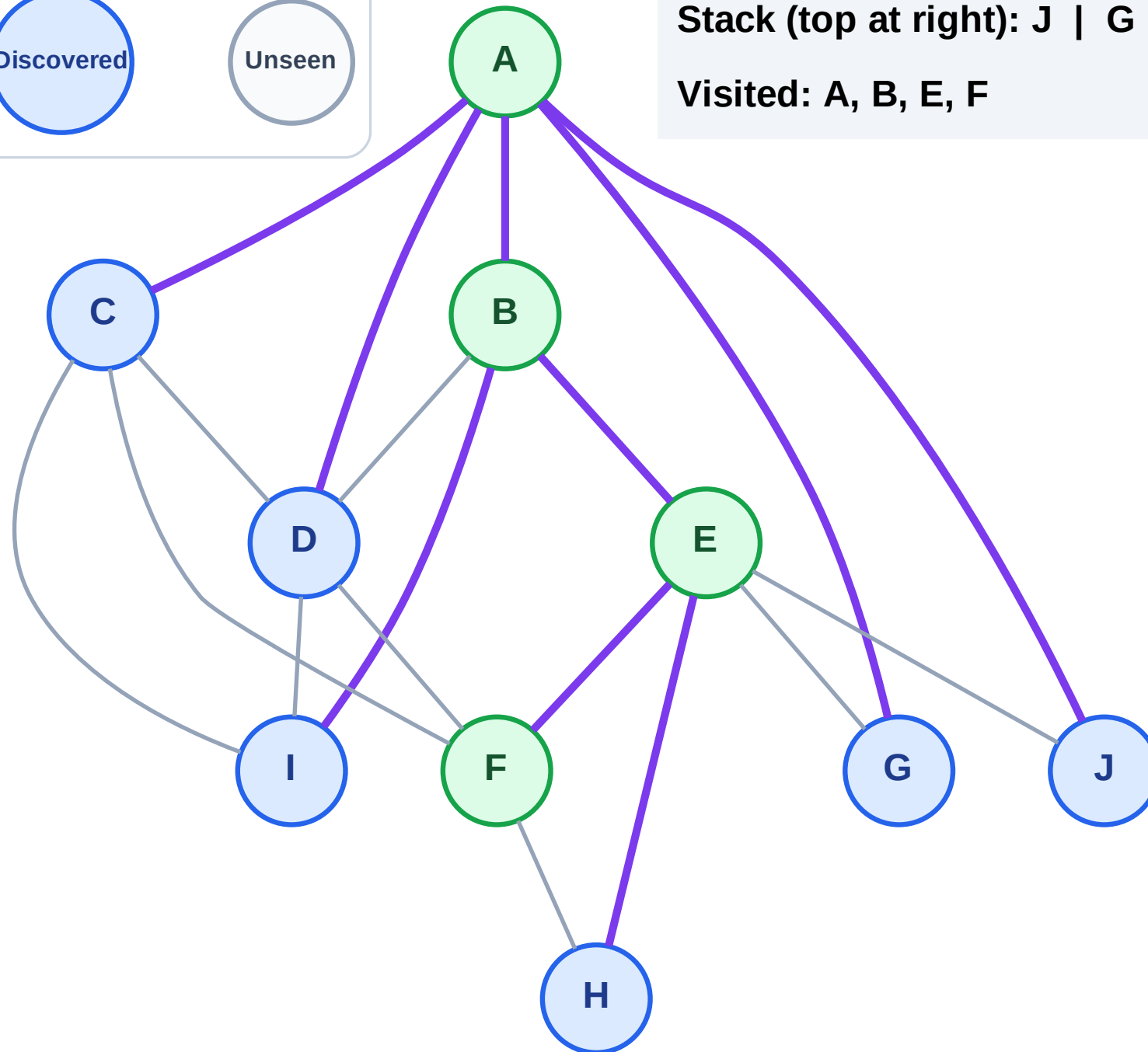
Step 9: No new neighbors from F

## Legend



Stack (top at right): J | G | D | C | I | H

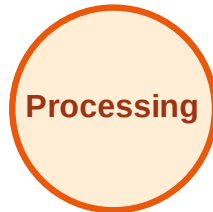
Visited: A, B, E, F



# DFS Traversal

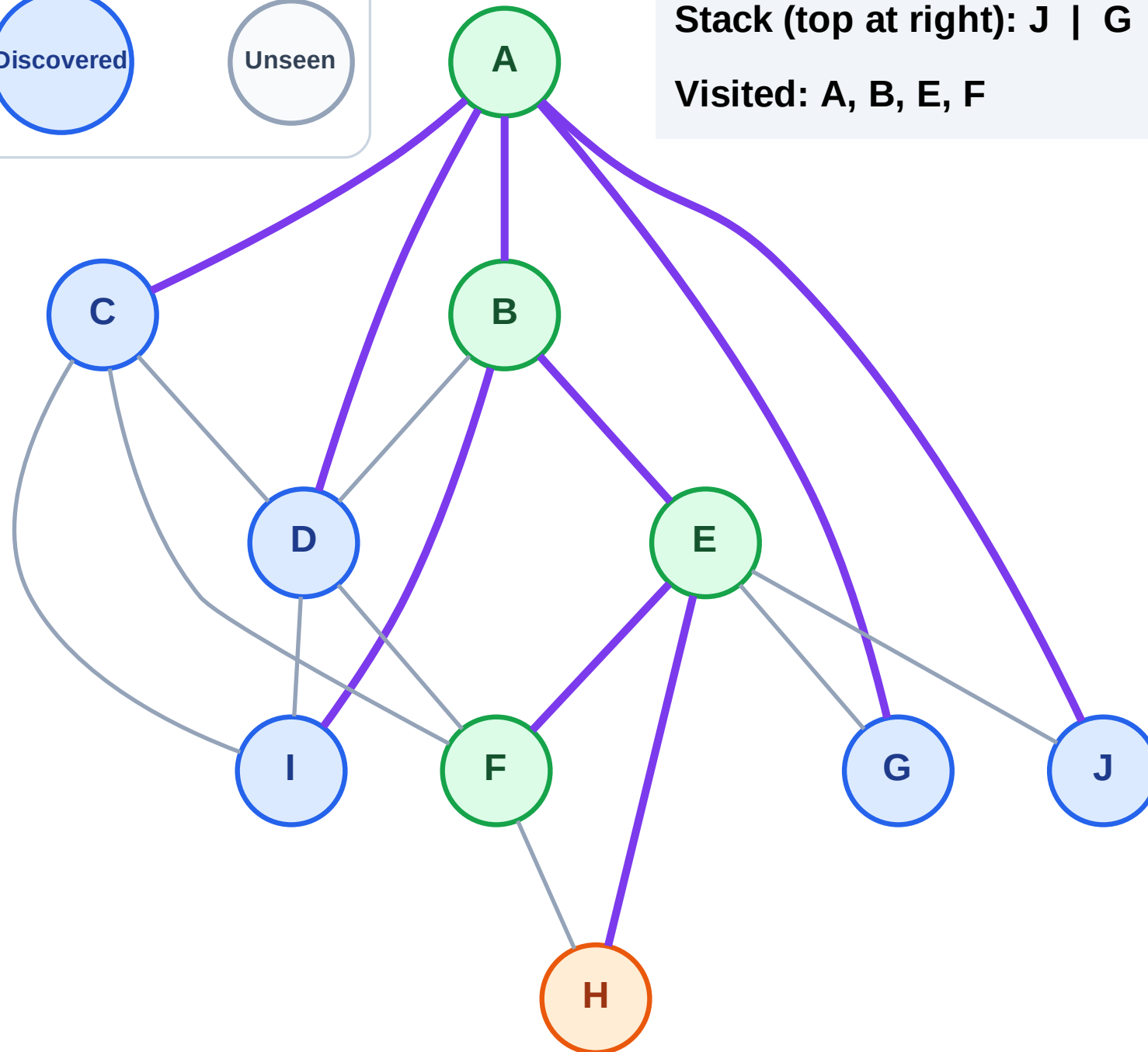
Step 10: Process node H

## Legend



Stack (top at right): J | G | D | C | I

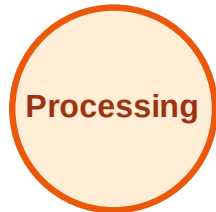
Visited: A, B, E, F



# DFS Traversal

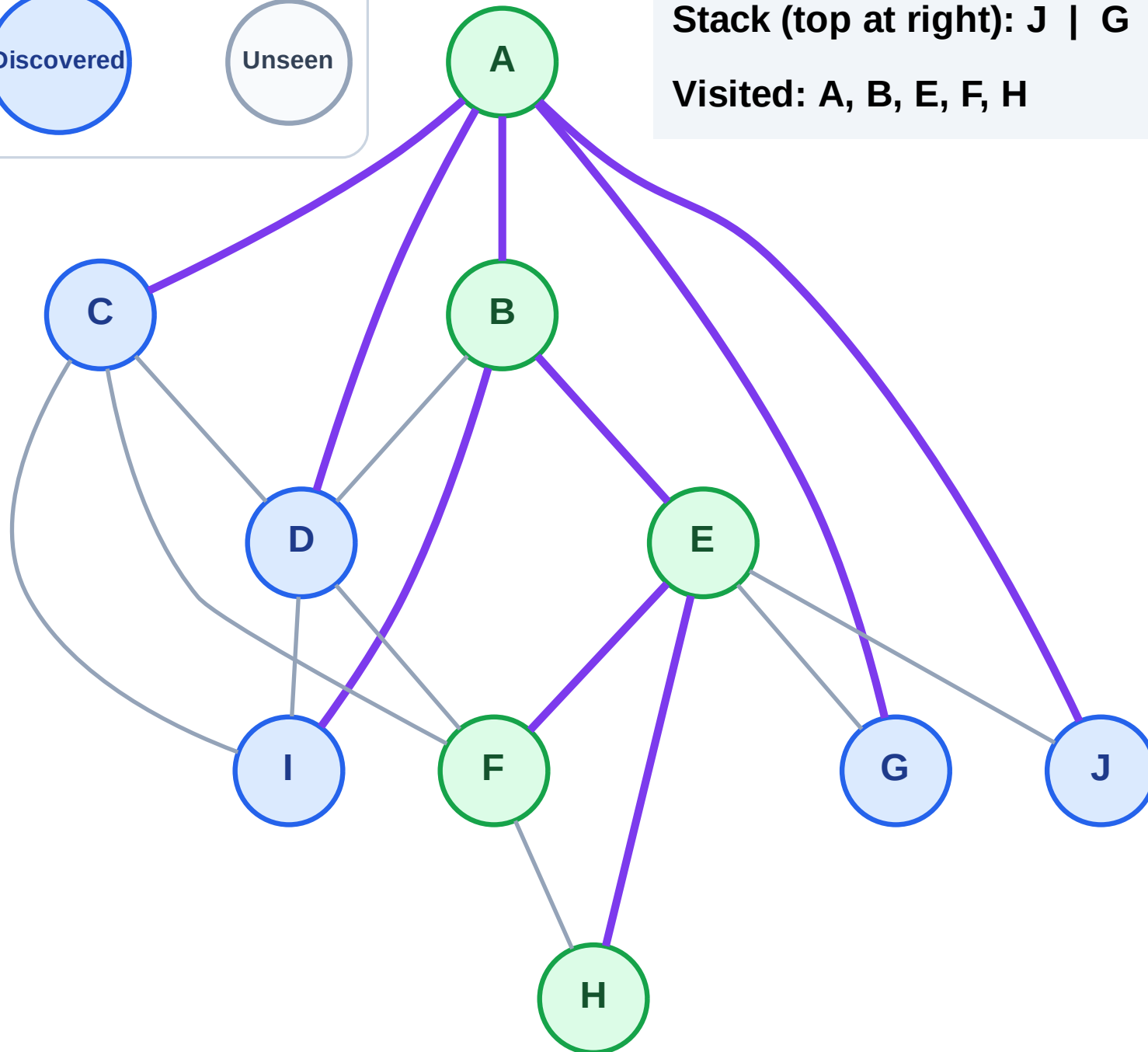
Step 11: No new neighbors from H

## Legend



Stack (top at right): J | G | D | C | I

Visited: A, B, E, F, H



# DFS Traversal

Step 12: Process node I

Legend

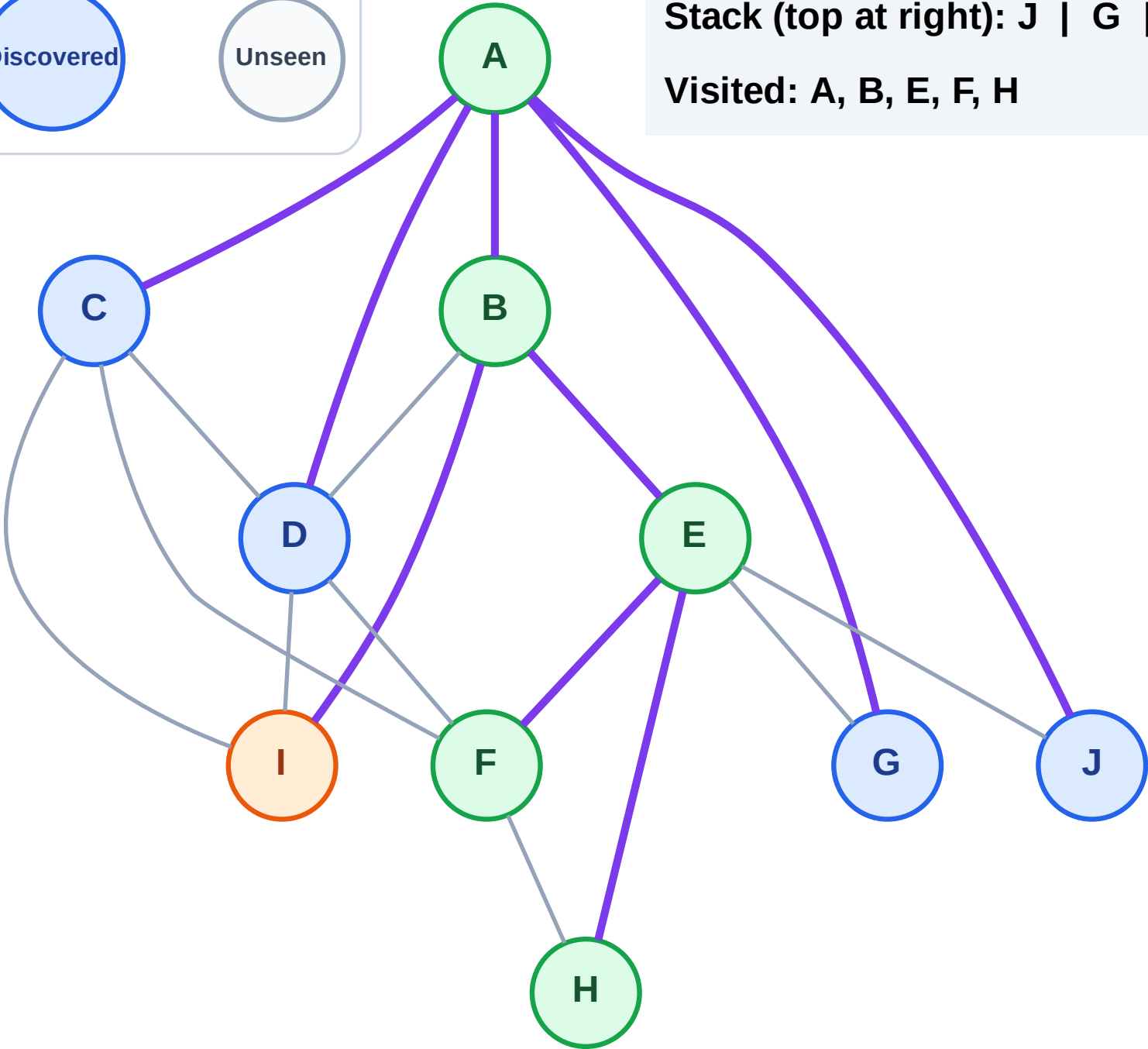
Complete

Processing

Discovered

Unseen

Stack (top at right): J | G | D | C  
Visited: A, B, E, F, H



# DFS Traversal

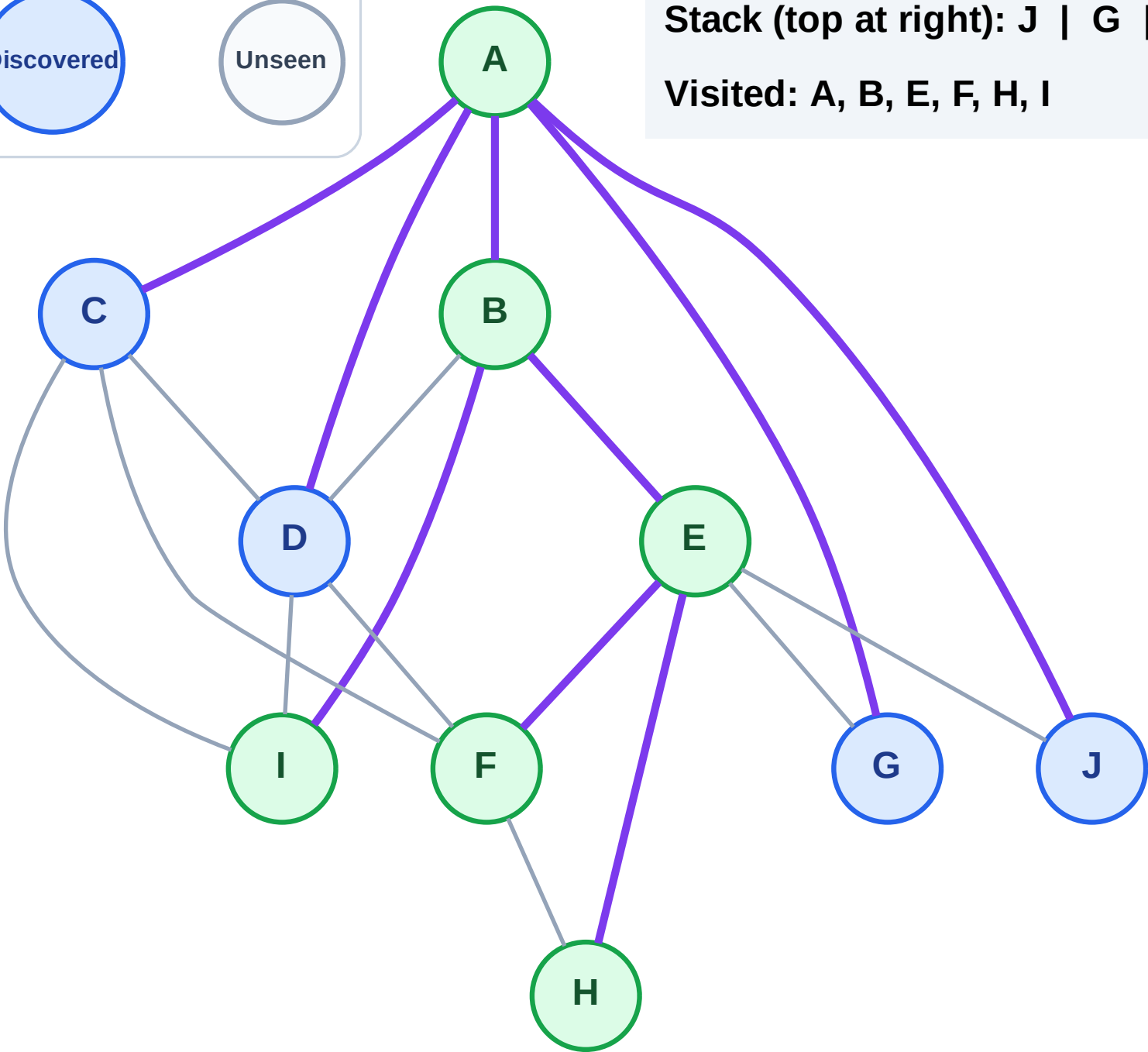
Step 13: No new neighbors from I

## Legend



Stack (top at right): J | G | D | C

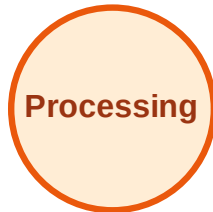
Visited: A, B, E, F, H, I



# DFS Traversal

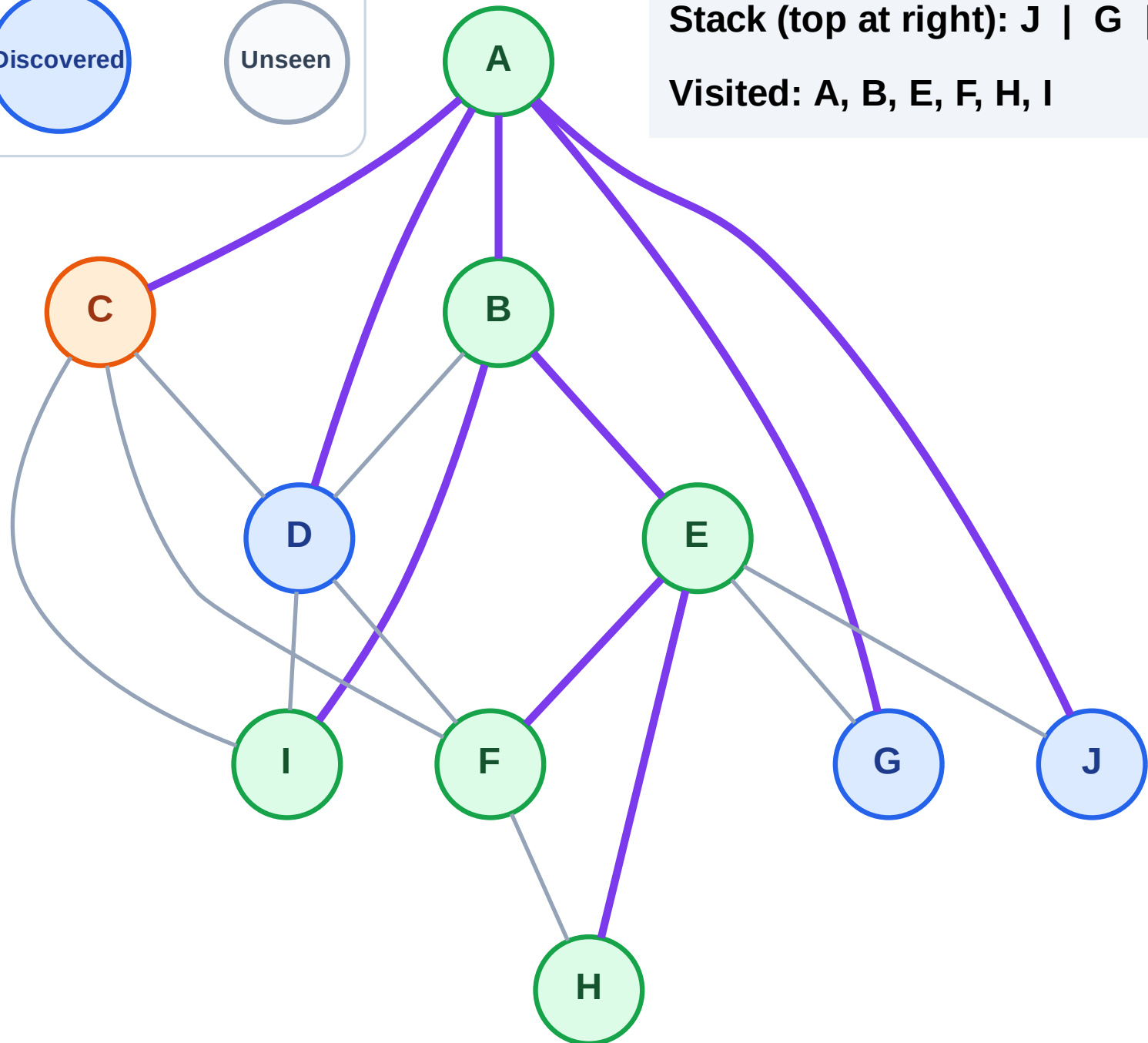
Step 14: Process node C

## Legend



Stack (top at right): J | G | D

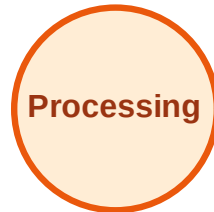
Visited: A, B, E, F, H, I



# DFS Traversal

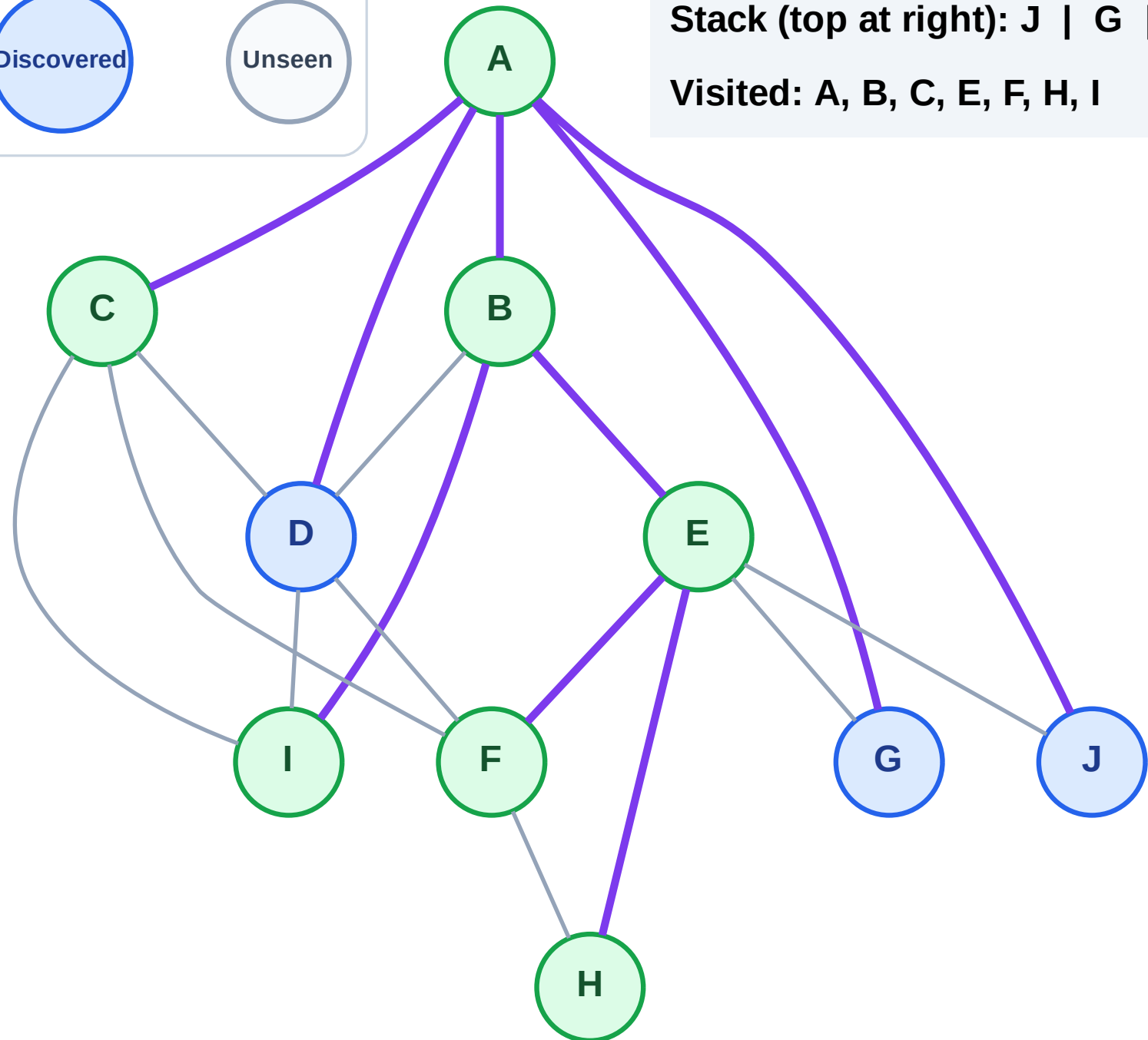
Step 15: No new neighbors from C

## Legend



Stack (top at right): J | G | D

Visited: A, B, C, E, F, H, I



# DFS Traversal

Step 16: Process node D

## Legend

Complete

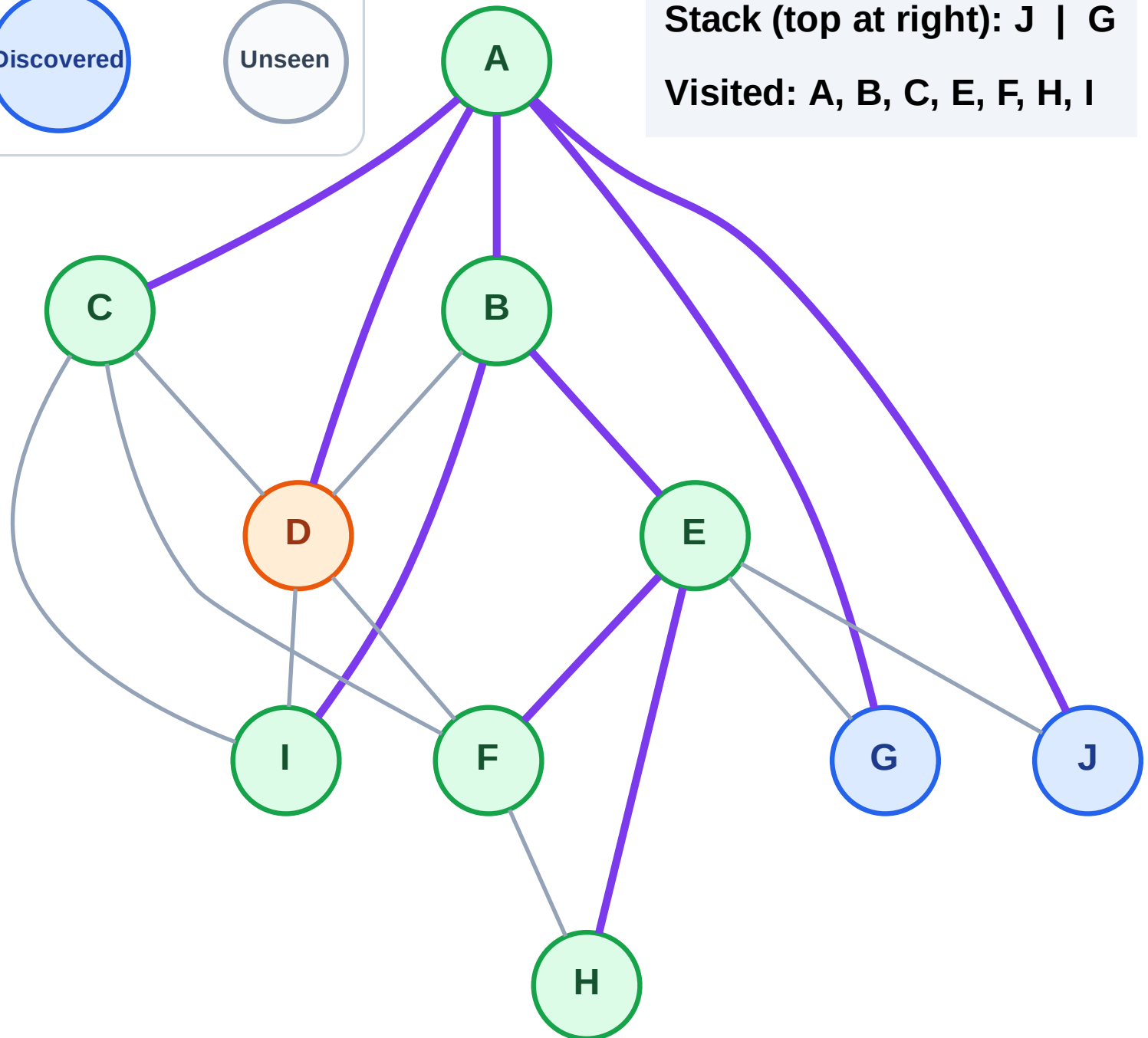
Processing

Discovered

Unseen

Stack (top at right): J | G

Visited: A, B, C, E, F, H, I



# DFS Traversal

Step 17: No new neighbors from D

## Legend

Complete

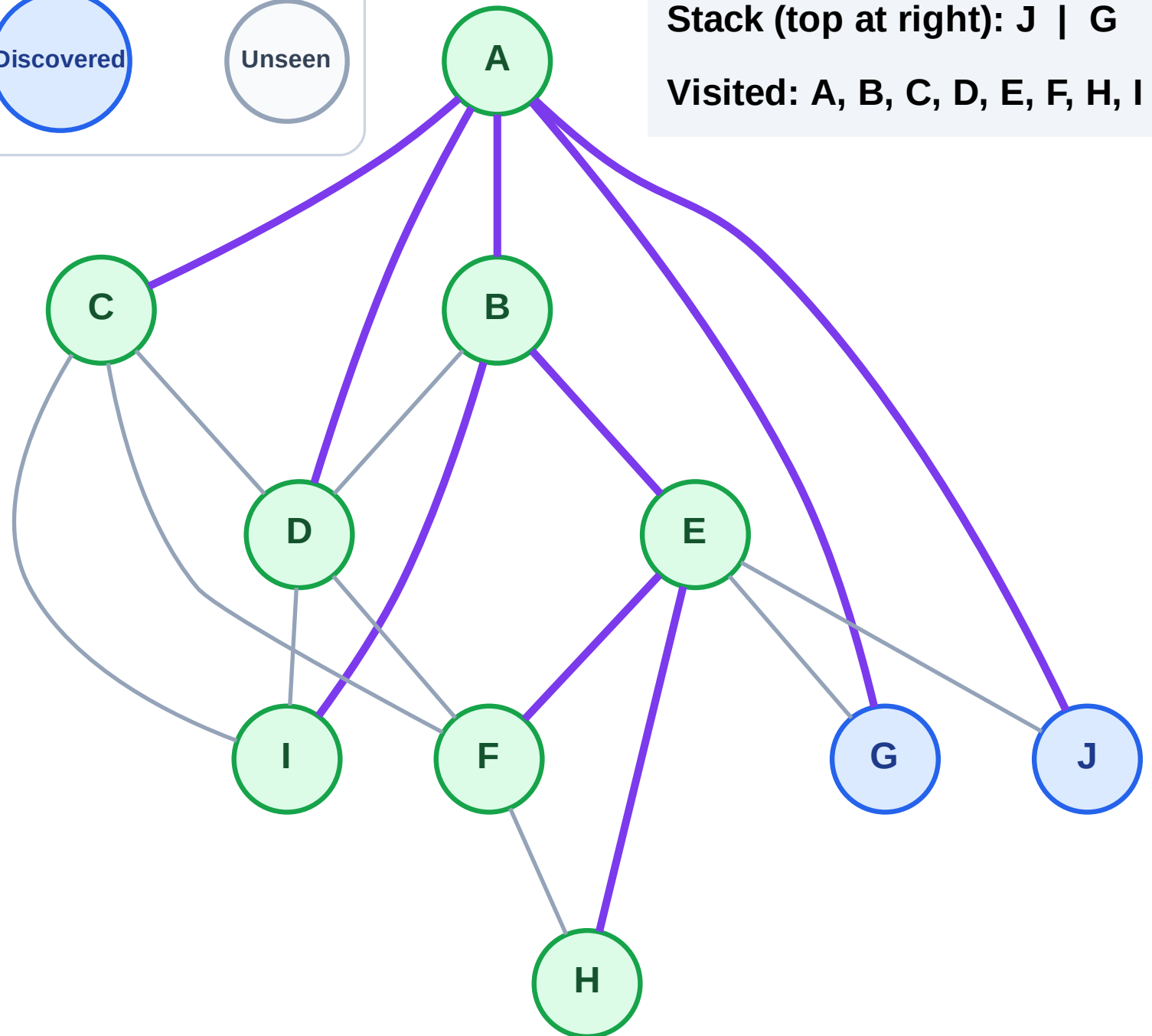
Processing

Discovered

Unseen

Stack (top at right): J | G

Visited: A, B, C, D, E, F, H, I



# DFS Traversal

Step 18: Process node G

## Legend

Complete

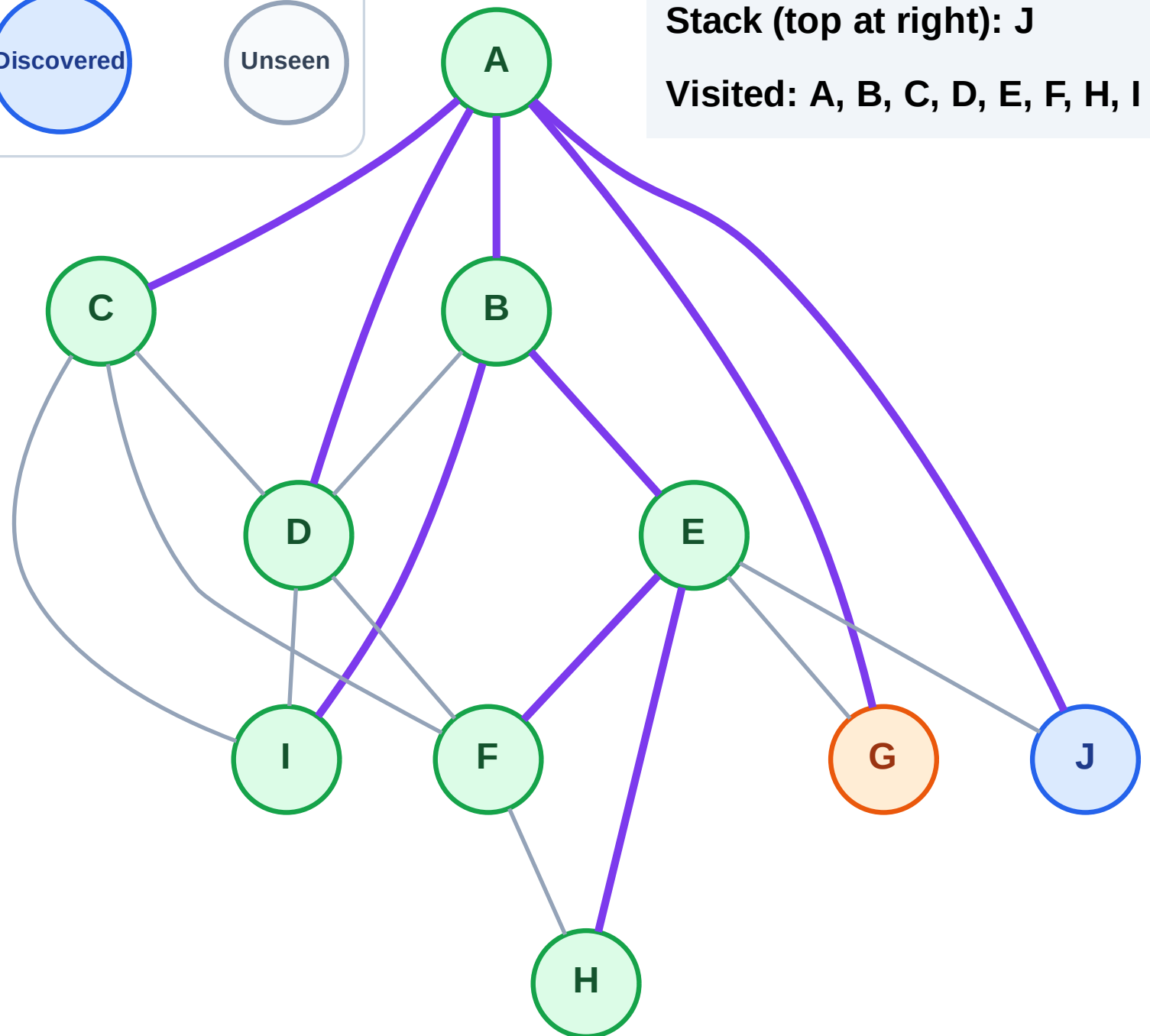
Processing

Discovered

Unseen

Stack (top at right): J

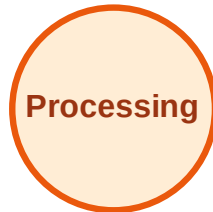
Visited: A, B, C, D, E, F, H, I



# DFS Traversal

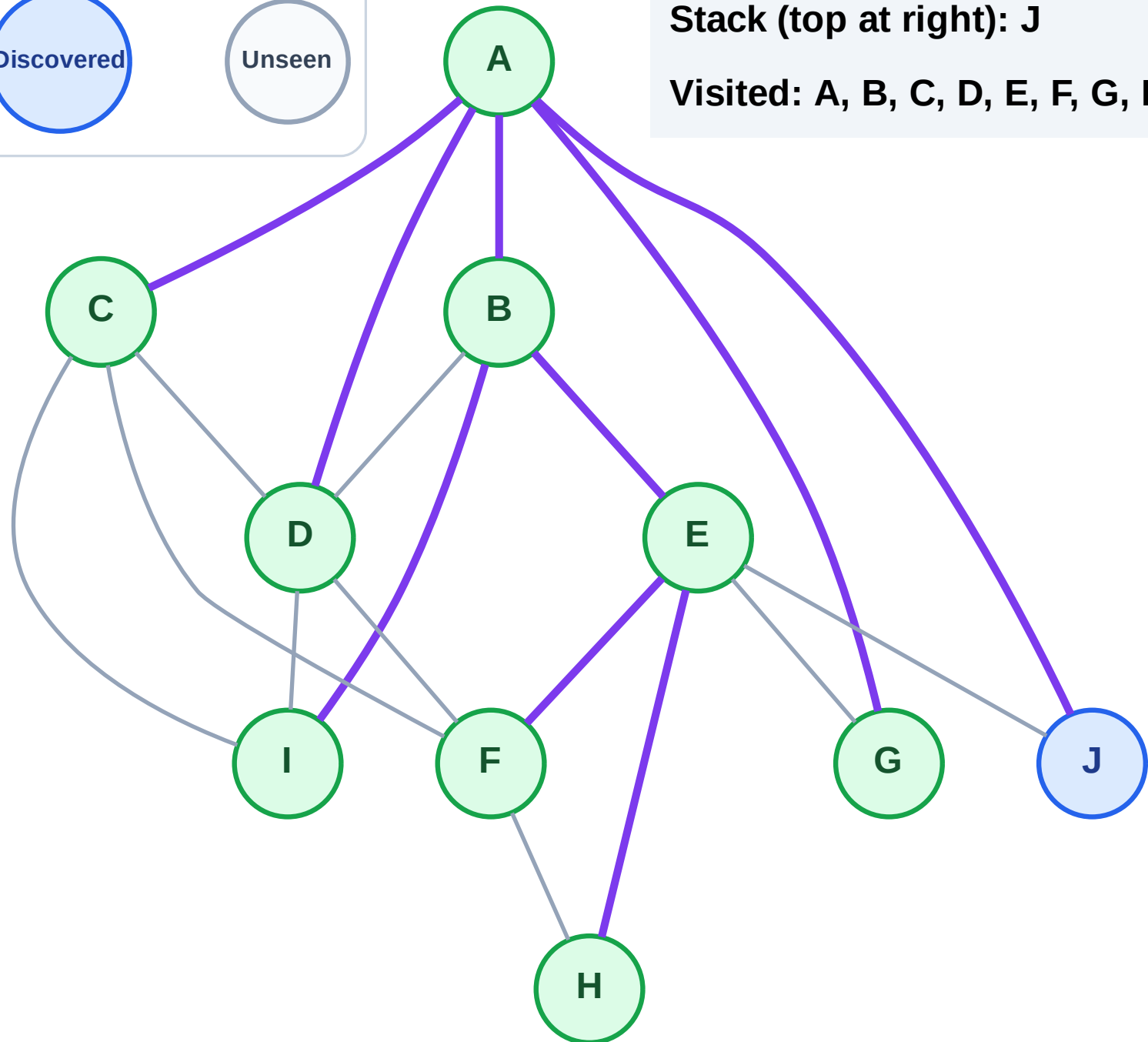
Step 19: No new neighbors from G

## Legend



Stack (top at right): J

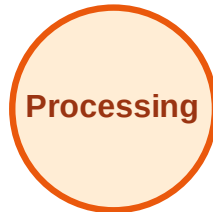
Visited: A, B, C, D, E, F, G, H, I



# DFS Traversal

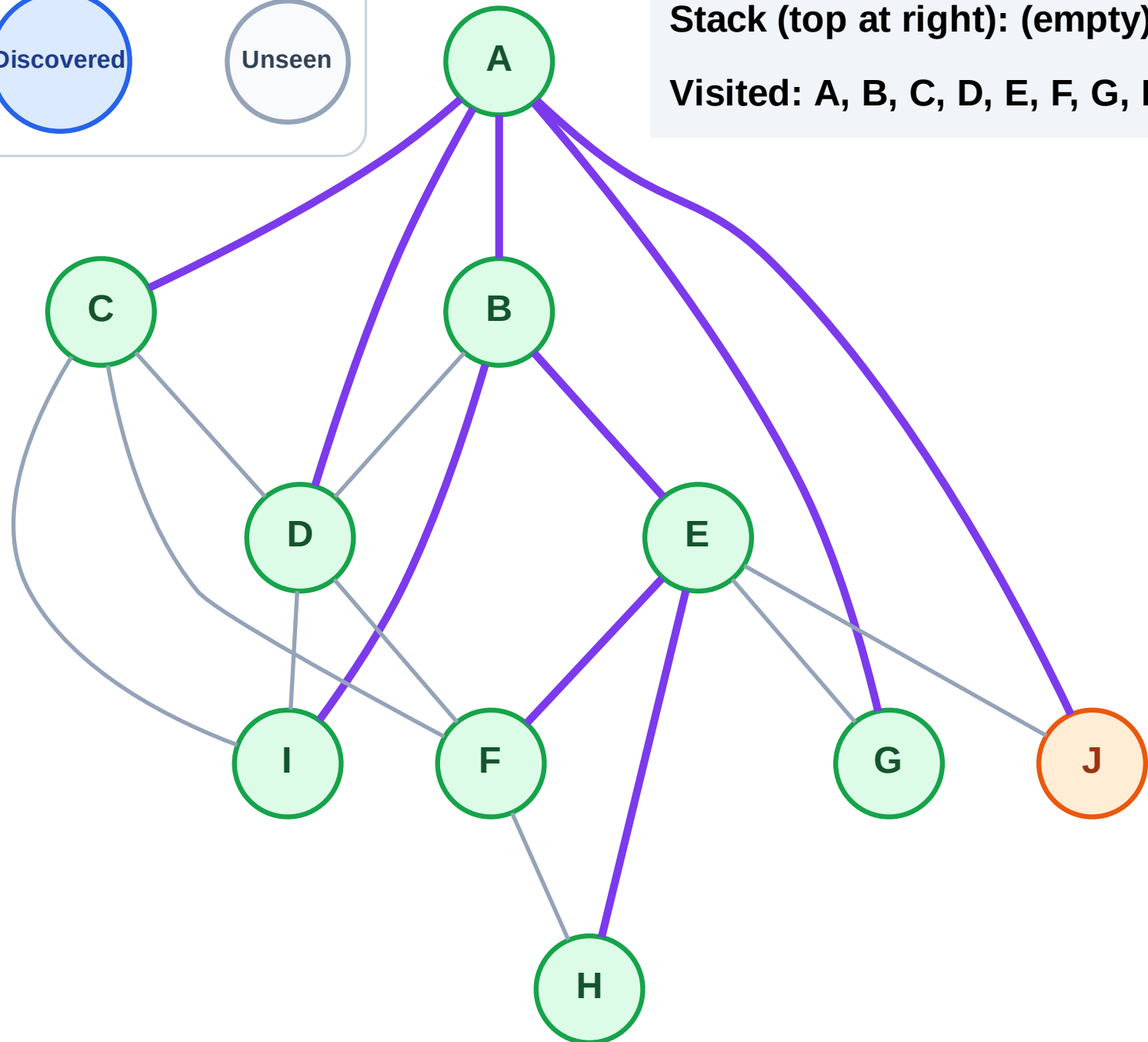
Step 20: Process node J

## Legend



Stack (top at right): (empty)

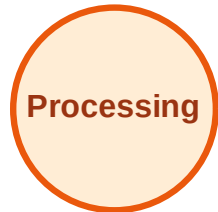
Visited: A, B, C, D, E, F, G, H, I



# DFS Traversal

Step 21: No new neighbors from J

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

